PS3

Problem 1

(A)

(A) $y = \frac{P}{6EI}(x^3 - 3Lx^2)$

$$\boxed{y(x) = \frac{x^3}{6} - \frac{x^2}{2}}$$

(B) $y = \frac{-wx^2}{24EI}(6L^2 - 4Lx + x^2)$

$$\boxed{y(x) = \frac{x^2}{24}(6 - 4x + x^2)}$$

(C) $y = \frac{-w}{24EI}(x^4 - 2Lx^3 + L^3x)$

$$\boxed{y(x) = -\frac{1}{24}(x^4 - 2x^3 + x)}$$

(B)

PINN for (A)

In [16]:
```
# Import TensorFlow and NumPy
import tensorflow as tf
import numpy as np
from pathlib import Path

# Set data type
DTYPE = "float32"
tf.keras.backend.set_floatx(DTYPE)

# Constants: EI = L = w = F = 1 (writeups use P for tip load; code uses F)
```

```
L = tf.constant(1.0, dtype=DTYPE)
F = tf.constant(1.0, dtype=DTYPE)  # tip force at x=L, scenario (A)
W = tf.constant(1.0, dtype=DTYPE)  # uniform load intensity w, scenarios (B) and (C

# PDE: f'''' = Q_load  (set before training: 0 for A, W for B/C with EI=1)
Q_load = tf.Variable(0.0, trainable=False, dtype=DTYPE)


def make_model():
    return tf.keras.Sequential(
        [
            tf.keras.layers.Input(shape=(1,)),
            tf.keras.layers.Dense(64, activation="tanh"),
            tf.keras.layers.Dense(64, activation="tanh"),
            tf.keras.layers.Dense(64, activation="tanh"),
            tf.keras.layers.Dense(64, activation="tanh"),
            tf.keras.layers.Dense(1),
        ]
    )


model = make_model()


def f_and_derivs(x):
    """Return f, f', f'', f''', f'''' at x (eager: must stay eager because `model`
    x = tf.convert_to_tensor(x)
    with tf.GradientTape(persistent=True) as t4:
        t4.watch(x)
        with tf.GradientTape(persistent=True) as t3:
            t3.watch(x)
            with tf.GradientTape(persistent=True) as t2:
                t2.watch(x)
                with tf.GradientTape() as t1:
                    t1.watch(x)
                    f = model(x)
                f_x = t1.gradient(f, x)
            f_xx = t2.gradient(f_x, x)
        f_xxx = t3.gradient(f_xx, x)
    f_xxxx = t4.gradient(f_xxx, x)
    del t2, t3, t4
    return f, f_x, f_xx, f_xxx, f_xxxx


def residual(x):
    _, _, _, _, f_xxxx = f_and_derivs(x)
    return f_xxxx - Q_load


def mean_square(tensor):
    return tf.reduce_mean(tf.square(tensor))


# Case (A): cantilever, tip load F at x=L (diagram P = F in code), q=0 in domain.
# Part (a) y = F/(6EI)(x^3 - 3Lx^2) gives EI y'''(L) = F at x=L (EI=1).
# If your grader insists on EI v'''(L) = -F from the figure, use shear_free + F ins
```

```python
def bc_terms_case_A():
    x_fixed = tf.zeros((1, 1), dtype=DTYPE)
    x_free = tf.ones((1, 1), dtype=DTYPE) * L
    f_fixed, slope_fixed, _, _, _ = f_and_derivs(x_fixed)
    _, _, moment_free, shear_free, _ = f_and_derivs(x_free)
    # Keys must not contain f''' inside "..." — TensorFlow AutoGraph misparses that
    return {
        "disp_0": f_fixed,
        "slope_0": slope_fixed,
        "moment_L": moment_free,
        "shear_L_minus_F": shear_free - F,
    }


# Case (B): cantilever, uniform load — f'''' = w; free end: M=V=0
def bc_terms_case_B():
    x_fixed = tf.zeros((1, 1), dtype=DTYPE)
    x_free = tf.ones((1, 1), dtype=DTYPE) * L
    f_fixed, slope_fixed, _, _, _ = f_and_derivs(x_fixed)
    _, _, moment_free, shear_free, _ = f_and_derivs(x_free)
    return {
        "disp_0": f_fixed,
        "slope_0": slope_fixed,
        "moment_L": moment_free,
        "shear_L": shear_free,
    }


# Case (C): simply supported, uniform load — f=M=0 at both ends
def bc_terms_case_C():
    x0 = tf.zeros((1, 1), dtype=DTYPE)
    xL = tf.ones((1, 1), dtype=DTYPE) * L
    f0, _, m0, _, _ = f_and_derivs(x0)
    fL, _, mL, _, _ = f_and_derivs(xL)
    return {
        "disp_0": f0,
        "disp_L": fL,
        "moment_0": m0,
        "moment_L": mL,
    }


def train_pinn(bc_terms_fn, num_steps=5000, print_every=200, lr=1e-3, N_col=200):
    """Collocation on (0, L); BC residuals from bc_terms_fn."""
    global model
    x_col = np.random.rand(N_col, 1).astype(np.float32) * float(L.numpy())
    x_col = tf.convert_to_tensor(x_col, dtype=DTYPE)
    optimizer = tf.keras.optimizers.Adam(lr)

    @tf.function
    def train_step(xc):
        with tf.GradientTape() as tape:
            r = residual(xc)
            loss_pde = mean_square(r)
            bc = bc_terms_fn()
            loss_bc = tf.constant(0.0, dtype=DTYPE)
```

```
            for name in bc:
                loss_bc = loss_bc + mean_square(bc[name])
            loss_total = loss_pde + loss_bc
        grads = tape.gradient(loss_total, model.trainable_variables)
        optimizer.apply_gradients(zip(grads, model.trainable_variables))
        return loss_total, loss_pde, loss_bc

    for step in range(num_steps):
        loss_total, loss_pde, loss_bc = train_step(x_col)
        if step % print_every == 0:
            print(
                "step:", step,
                "  total:", float(loss_total.numpy()),
                "  pde:", float(loss_pde.numpy()),
                "  bc:", float(loss_bc.numpy()),
            )


def local_homework_dir():
    for p in [Path(".").resolve()] + list(Path(".").resolve().parents):
        if (p / "PS3.ipynb").is_file():
            return p
    return Path(".").resolve()


def f_analytic_A(x):
    """Part (a): y = F/(6EI) (x^3 - 3 L x^2); with EI=L=F=1 this is x^3/6 - x^2/2."
    x = np.asarray(x, dtype=np.float64)
    Lv, Fv = float(L.numpy()), float(F.numpy())
    return (Fv / 6.0) * (x ** 3 - 3.0 * Lv * x ** 2)


def f_analytic_B(x):
    """Part (a): y = (w x^2)/(24 EI) (6 L^2 - 4 L x + x^2); L=w=EI=1 matches homewo
    x = np.asarray(x, dtype=np.float64)
    Lv, wv = float(L.numpy()), float(W.numpy())
    return (wv / 24.0) * x ** 2 * (6.0 * Lv ** 2 - 4.0 * Lv * x + x ** 2)


def f_analytic_C(x):
    """Part (a): y = -w/(24 EI) (x^4 - 2 L x^3 + L^3 x); L=w=EI=1 matches homework
    x = np.asarray(x, dtype=np.float64)
    Lv, wv = float(L.numpy()), float(W.numpy())
    return -(wv / 24.0) * (x ** 4 - 2.0 * Lv * x ** 3 + (Lv ** 3) * x)


def plot_pinn_vs_analytic(
    title,
    f_analytic_fn,
    save_name,
    *,
    invert_yaxis=True,
    analytic_eq=None,
):
    import matplotlib.pyplot as plt
```

```
    Lv = float(L.numpy())
    x_plot = np.linspace(0.0, Lv, 200).reshape(-1, 1).astype(np.float32)
    x_tf = tf.convert_to_tensor(x_plot, dtype=DTYPE)
    f_pinn = model(x_tf).numpy().ravel()
    f_true = f_analytic_fn(x_plot.ravel())

    charts_dir = local_homework_dir() / "charts"
    charts_dir.mkdir(parents=True, exist_ok=True)

    fig, ax = plt.subplots(figsize=(8, 5))
    ax.plot(x_plot.ravel(), f_true, "k-", label="Analytic")
    ax.plot(x_plot.ravel(), f_pinn, "r--", label="PINN")
    if invert_yaxis:
        ax.invert_yaxis()
    ax.set_xlabel("x (along beam)")
    ax.set_ylabel("deflection f(x)")
    ax.set_title(title)
    ax.legend(loc="lower right")
    ax.grid(True, alpha=0.3)
    if analytic_eq:
        ax.text(
            0.02,
            0.5,
            analytic_eq,
            transform=ax.transAxes,
            fontsize=11,
            verticalalignment="center",
            bbox=dict(boxstyle="round,pad=0.35", facecolor="wheat", alpha=0.9),
        )
    out = charts_dir / save_name
    fig.savefig(out, dpi=150, bbox_inches="tight")
    plt.show()
    print("Saved:", out.resolve())
```

In [17]:

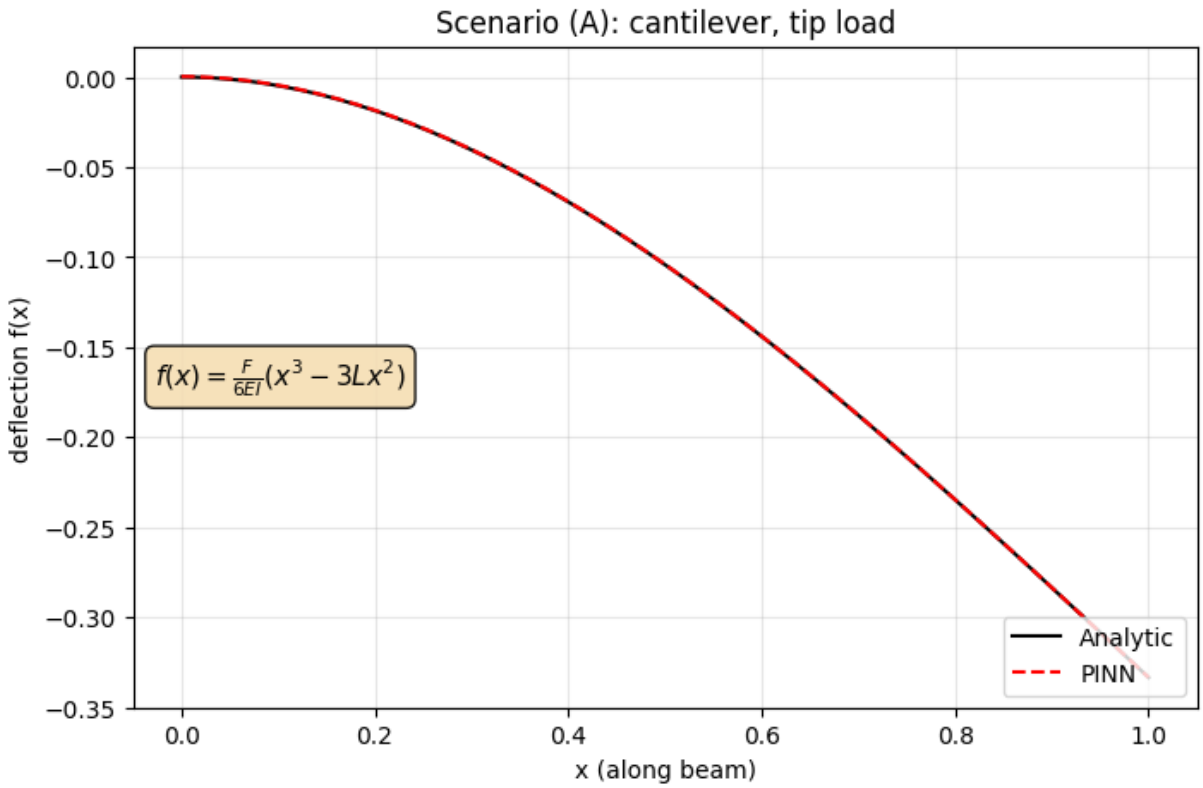
```
# Scenario (A): cantilever, tip load — f'''' = 0 in domain
Q_load.assign(0.0)
model = make_model()
train_pinn(bc_terms_case_A)
```

```
step: 0   total: 1.0190595388412476   pde: 0.0009086098289117217   bc: 1.01815092563
62915
step: 200   total: 0.0007133532781153917   pde: 0.0005998734850436449   bc: 0.000113
47980034770444
step: 400   total: 0.0006066995556466281   pde: 0.00045115090324543417   bc: 0.00015
554865240119398
step: 600   total: 4.749956860905513e-05   pde: 3.6620480386773124e-05   bc: 1.08790
86403292604e-05
step: 800   total: 2.8000164093100466e-05   pde: 2.691403278731741e-05   bc: 1.08613
1987904082e-06
step: 1000   total: 2.36538726312574e-05   pde: 2.3634540411876515e-05   bc: 1.93317
09566472455e-08
step: 1200   total: 5.9770700318040326e-05   pde: 4.4847867684438825e-05   bc: 1.492
2832633601502e-05
step: 1400   total: 2.0341391063993797e-05   pde: 1.9716839233296923e-05   bc: 6.245
512622626848e-07
step: 1600   total: 0.0024317384231835604   pde: 0.001965239178389311   bc: 0.000466
4993321057409
step: 1800   total: 0.0001836946321418509   pde: 2.1351448594941758e-05   bc: 0.0001
6234318900387734
step: 2000   total: 1.3632391528517473e-05   pde: 1.3548420611186884e-05   bc: 8.397
0775222042e-08
step: 2200   total: 2.758675691438839e-05   pde: 1.654130028327927e-05   bc: 1.10454
5754060382e-05
step: 2400   total: 0.0002003302361117676   pde: 4.9378519179299474e-05   bc: 0.0001
5095171693246812
step: 2600   total: 0.0001427782408427447   pde: 9.377617243444547e-05   bc: 4.90020
6477032043e-05
step: 2800   total: 4.434079892234877e-05   pde: 1.5181438357103616e-05   bc: 2.9159
36056524515e-05
step: 3000   total: 8.269740646937862e-06   pde: 8.122189683490433e-06   bc: 1.47550
79291717266e-07
step: 3200   total: 0.0015883214073255658   pde: 0.0009552774718031287   bc: 0.00063
30439355224371
step: 3400   total: 0.0033410959877073765   pde: 0.0013097560731694102   bc: 0.00203
13397981226444
step: 3600   total: 6.649544957326725e-06   pde: 6.10547022006358e-06   bc: 5.440745
098894695e-07
step: 3800   total: 9.596922609489411e-06   pde: 7.036335773591418e-06   bc: 2.56058
70632716687e-06
step: 4000   total: 0.0003871625813189894   pde: 2.300228879903443e-05   bc: 0.00036
416028160601854
step: 4200   total: 4.189916126051685e-06   pde: 4.184303634247044e-06   bc: 5.61235
6801520946e-09
step: 4400   total: 3.7914708173047984e-06   pde: 3.763219638130977e-06   bc: 2.8251
232464526765e-08
step: 4600   total: 9.853061783360317e-05   pde: 3.1663570553064346e-05   bc: 6.6867
04728053883e-05
step: 4800   total: 0.000886776833795011   pde: 0.00041390082333236933   bc: 0.00047
28760104626417
```

In [18]:

```python
# Scenario (A): PINN vs analytic, save under charts/
plot_pinn_vs_analytic(
    "Scenario (A): cantilever, tip load",
    f_analytic_A,
    "1b_A.png",
```

```
    invert_yaxis=False,
    analytic_eq=r"$f(x)=\frac{F}{6EI}\left(x^3-3Lx^2\right)$",
)
bc = bc_terms_case_A()
print("Boundary condition residuals (A):")
for name in bc:
    print(" ", name, "=", bc[name].numpy().reshape(-1))
```




```
Saved: C:\Users\mpave\OneDrive - Massachusetts Institute of Technology\2.174 Mechani
cs and Materials via Machine Learning\PS3\charts\1b_A.png
Boundary condition residuals (A):
  disp_0 = [0.00027183]
  slope_0 = [-0.00025749]
  moment_L = [2.4862587e-05]
  shear_L_minus_F = [4.4822693e-05]
```

PINN for (B)

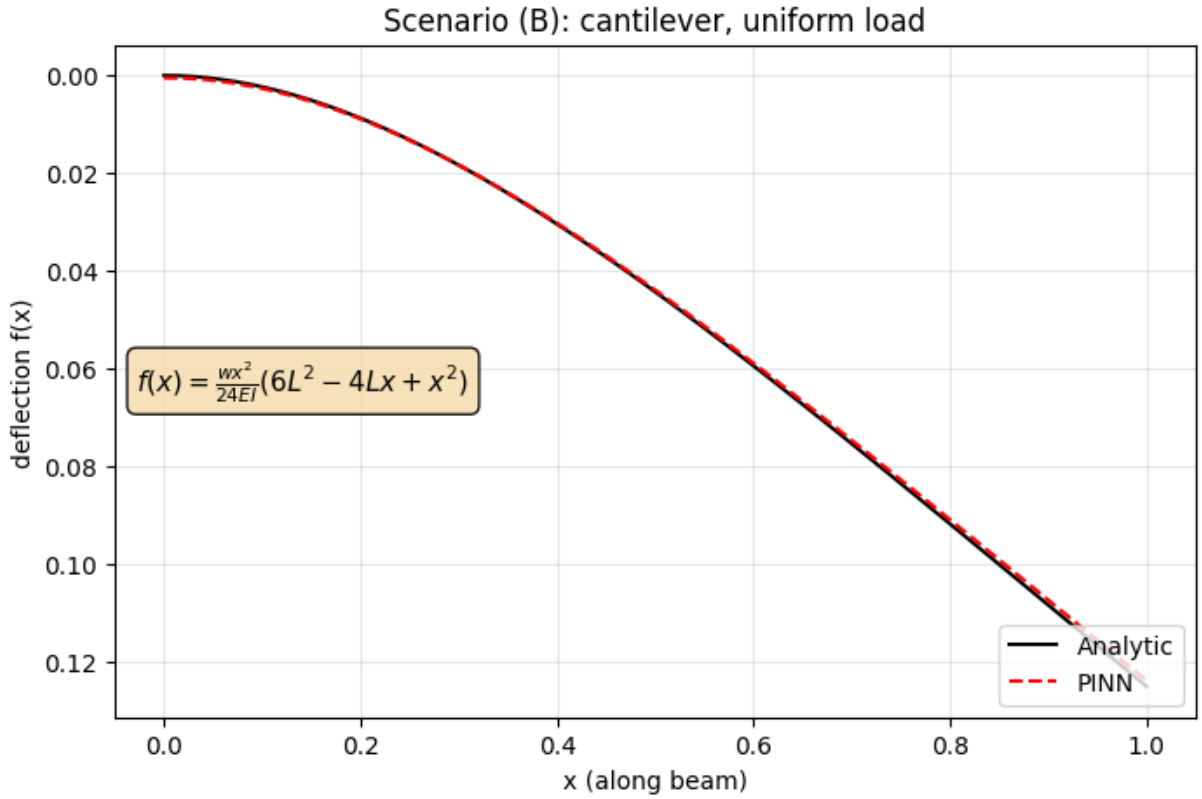
In [19]:
```
# Scenario (B): cantilever, uniform load — f'''' = w (EI=1 → Q_load = W)
Q_load.assign(W)
model = make_model()
train_pinn(bc_terms_case_B)
```

```
step: 0   total: 0.5899721384048462   pde: 0.5486840605735779   bc: 0.04128809645771
98
step: 200   total: 0.00837371964007616   pde: 0.006702465936541557   bc: 0.001671253
5871192813
step: 400   total: 0.0030004503205418587   pde: 0.0029078256338834763   bc: 9.262465
755455196e-05
step: 600   total: 0.001602465519681573   pde: 0.0015847162576392293   bc: 1.7749205
653672107e-05
step: 800   total: 0.000927821034565568   pde: 0.0009227703558281064   bc: 5.0506514
52620514e-06
step: 1000   total: 0.0005342842196114361   pde: 0.0005327160470187664   bc: 1.56818
96456953837e-06
step: 1200   total: 0.00021669188572559506   pde: 0.00021223870862741023   bc: 4.453
173005458666e-06
step: 1400   total: 4.206388985039666e-05   pde: 4.013189027318731e-05   bc: 1.93199
775821995e-06
step: 1600   total: 1.961556881724391e-05   pde: 1.925087235576939e-05   bc: 3.64695
7793534966e-07
step: 1800   total: 1.5256876395142172e-05   pde: 1.5087272004166152e-05   bc: 1.696
043767651645e-07
step: 2000   total: 1.4327559256344102e-05   pde: 1.4208676475391258e-05   bc: 1.188
8301543194757e-07
step: 2200   total: 1.0480390301381703e-05   pde: 1.0473046131664887e-05   bc: 7.344
48235562013e-09
step: 2400   total: 2.991062319779303e-05   pde: 2.7868323741131462e-05   bc: 2.0422
99456661567e-06
step: 2600   total: 0.0001741206069709733   pde: 0.0001594251807546243   bc: 1.46954
31673317216e-05
step: 2800   total: 1.1738821740436833e-05   pde: 1.0868771823879797e-05   bc: 8.700
500302438741e-07
step: 3000   total: 8.082270142040215e-06   pde: 8.07560354587622e-06   bc: 6.666373
231212219e-09
step: 3200   total: 9.731524914968759e-05   pde: 2.094338196911849e-05   bc: 7.63718
708185479e-05
step: 3400   total: 7.355198249570094e-06   pde: 7.351920885412255e-06   bc: 3.27737
2595533734e-09
step: 3600   total: 1.54238186951261e-05   pde: 1.3311259863257874e-05   bc: 2.11255
92866155785e-06
step: 3800   total: 1.141194297815673e-05   pde: 9.706153832667042e-06   bc: 1.70578
8690742338e-06
step: 4000   total: 7.88026227382943e-06   pde: 7.834266398276668e-06   bc: 4.599584
002562551e-08
step: 4200   total: 1.8332295439904556e-05   pde: 1.534898729005363e-05   bc: 2.9833
088319719536e-06
step: 4400   total: 7.205123893072596e-06   pde: 7.201795142464107e-06   bc: 3.32870
7531835562e-09
step: 4600   total: 7.48936872696504e-05   pde: 1.1201603228983004e-05   bc: 6.36920
8131218329e-05
step: 4800   total: 0.0002412989706499502   pde: 2.4619605028419755e-05   bc: 0.0002
1667937107849866
```

In [20]:

```
plot_pinn_vs_analytic(
    "Scenario (B): cantilever, uniform load",
    f_analytic_B,
    "1b_B.png",
    analytic_eq=r"$f(x)=\frac{w x^2}{24EI}\left(6L^2-4Lx+x^2\right)$",
```

```
)
bc = bc_terms_case_B()
print("Boundary condition residuals (B):")
for name in bc:
    print(" ", name, "=", bc[name].numpy().reshape(-1))
```




```
Saved: C:\Users\mpave\OneDrive - Massachusetts Institute of Technology\2.174 Mechani
cs and Materials via Machine Learning\PS3\charts\1b_B.png
Boundary condition residuals (B):
  disp_0 = [0.00050299]
  slope_0 = [-0.00166413]
  moment_L = [0.00040685]
  shear_L = [0.00011769]
```

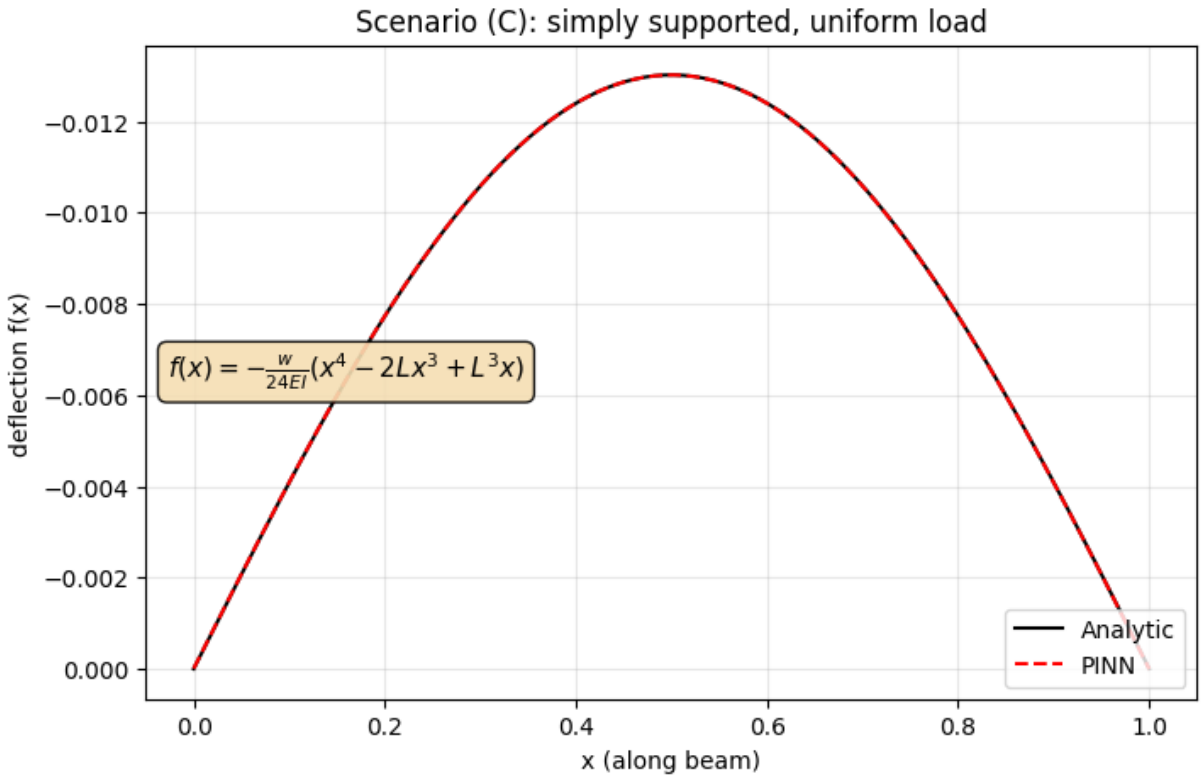
PINN for (C)

In [21]:
```
# Scenario (C): simply supported, uniform load w (code: W). Part (a) deflection has
Q_load.assign(-W)
model = make_model()
train_pinn(bc_terms_case_C)
```

```
step: 0   total: 0.9609578251838684   pde: 0.9584650993347168   bc: 0.00249271816574
03708
step: 200   total: 0.04262430965900421   pde: 0.03710347041487694   bc: 0.0055208406
411111355
step: 400   total: 0.0006170193664729595   pde: 0.0006152210989966989   bc: 1.798274
7522182763e-06
step: 600   total: 0.0004600304237101227   pde: 0.0004342392785474658   bc: 2.579115
2438614517e-05
step: 800   total: 0.0002635960699990392   pde: 0.00026295651332475245   bc: 6.39560
9943865566e-07
step: 1000   total: 0.00020265066996216774   pde: 0.00020215165568515658   bc: 4.990
158117834653e-07
step: 1200   total: 0.00010792318789754063   pde: 9.711916209198534e-05   bc: 1.0804
026715049986e-05
step: 1400   total: 7.609151361975819e-05   pde: 7.59352624299936e-05   bc: 1.562500
3868535714e-07
step: 1600   total: 2.0395047613419592e-05   pde: 1.793953742890153e-05   bc: 2.4555
097297707107e-06
step: 1800   total: 1.5909272406133823e-05   pde: 1.5876545148785226e-05   bc: 3.272
7477616845135e-08
step: 2000   total: 0.0009764712303876877   pde: 4.3144951632712036e-05   bc: 0.0009
333262569271028
step: 2200   total: 2.612388016132172e-06   pde: 2.6096656711160904e-06   bc: 2.7222
84619949278e-09
step: 2400   total: 1.3008943824388552e-05   pde: 1.298119968851097e-05   bc: 2.7744
469832668983e-08
step: 2600   total: 1.8694472601055168e-06   pde: 1.8685917666516616e-06   bc: 8.554
877362065838e-10
step: 2800   total: 8.766674000071362e-06   pde: 8.748657819523942e-06   bc: 1.80160
54426084338e-08
step: 3000   total: 1.7069903606170556e-06   pde: 1.7065692645701347e-06   bc: 4.210
545523353204e-10
step: 3200   total: 4.750856533064507e-06   pde: 4.721414370578714e-06   bc: 2.94419
43993901987e-08
step: 3400   total: 4.981135134585202e-05   pde: 2.07224929908989e-05   bc: 2.908886
017394252e-05
step: 3600   total: 5.8515033742878586e-05   pde: 9.895235052681528e-06   bc: 4.8619
796871207654e-05
step: 3800   total: 2.213566858699778e-06   pde: 2.212641902588075e-06   bc: 9.24904
985932784e-10
step: 4000   total: 2.036361593127367e-06   pde: 2.0358268102427246e-06   bc: 5.3487
55327538868e-10
step: 4200   total: 0.0002044487773673609   pde: 0.00019073294242843986   bc: 1.3715
829481952824e-05
step: 4400   total: 7.595740498800296e-06   pde: 4.147366780671291e-06   bc: 3.44837
37181290053e-06
step: 4600   total: 2.1734043912147172e-05   pde: 1.8912905943579972e-05   bc: 2.821
138423314551e-06
step: 4800   total: 3.1237582334142644e-06   pde: 2.8564986678247806e-06   bc: 2.672
5967927632155e-07
```

In [22]:
```
plot_pinn_vs_analytic(
    "Scenario (C): simply supported, uniform load",
    f_analytic_C,
    "1b_C.png",
    analytic_eq=r"$f(x)=-\frac{w}{24EI}\left(x^4-2Lx^3+L^3x\right)$",
```

```
)
bc = bc_terms_case_C()
print("Boundary condition residuals (C):")
for name in bc:
    print(" ", name, "=", bc[name].numpy().reshape(-1))
```




```
Saved: C:\Users\mpave\OneDrive - Massachusetts Institute of Technology\2.174 Mechani
cs and Materials via Machine Learning\PS3\charts\1b_C.png
Boundary condition residuals (C):
  disp_0 = [-8.667819e-06]
  disp_L = [7.3239207e-06]
  moment_0 = [-9.285286e-06]
  moment_L = [3.7446618e-05]
```

(C)

(i)

(c) $\boxed{\sigma = -\frac{M(x)y}{I}}$ $\tau = \frac{V(x)Q(x)}{Ib}$

$I = \frac{bh^3}{12} = \frac{1}{48}$ $A = bh = 0.5$

σ_{max} occurs at surface of beam $\rightarrow \frac{h}{2}$

$$\sigma_{max} = \frac{M(x)\left(\frac{h}{2}\right)}{\frac{1}{48}} = \frac{\frac{1}{2}M(x)}{\frac{1}{48}}$$

$$\boxed{\tau_{x,y} = \frac{3V(x)}{2A}\left(1 - \frac{4y^2}{h^2}\right)} \rightarrow$$ max shear stress at neutral axis

$$\left.\tau_{max} = \frac{3V(x)}{2A}\right\} \begin{matrix} A = 0.25 \\ h = 1 \end{matrix} \rightarrow \tau_{max} = 6V(x)$$

(ii)

(ii)

max @ $\frac{w}{2}$ max at surface

$$\sigma_{max} = \frac{M\left(\frac{y}{2}\right)}{I} = \frac{M\left(\frac{1}{2}\right)}{1/48} = \frac{\left(\frac{w}{2}\right)\left(\frac{1}{2}\right)}{1/48} = 12w$$

other stress is shear stress, which is greater?

$$\tau_{max} = \frac{3V}{2bh} = \frac{3V}{2A} = \frac{3w}{2(0.5)} = 6w$$

the max occurs at neutral axis

$$\sigma_{VM} = \sqrt{\sigma_{xx}^2 + 3\tau_{xy}^2} =$$

At surface $\rightarrow \tau = 0;\ \sigma = 12w$

$$10 = \sqrt{(12w)^2 + 0}$$
$$10 = 12w$$
$$\boxed{w = 0.83\bar{3}}$$

At neutral axis $\rightarrow \tau = 6w,\ \sigma = 0$

$$10 = \sqrt{0 + 3(6w)^2}$$
$$10 = \sqrt{3(36w^2)}$$
$$10 = \sqrt{108w^2}$$
$$10 = 10.39w$$
$$w = 0.96$$

Problem 2

(A)

In [30]:
```
# --- Problem 2(a): imports, paths, device ---
import os
from pathlib import Path

import numpy as np
```

```python
import pandas as pd
import torch
import torch.nn.functional as F
from torch.nn import Linear
from torch.optim.lr_scheduler import ReduceLROnPlateau
from sklearn.model_selection import train_test_split
from torch_geometric.data import Data
from torch_geometric.loader import DataLoader
from torch_geometric.nn import GCNConv, global_mean_pool

from tqdm.auto import tqdm


def homework_dir() -> Path:
    """Folder that contains PS3.ipynb (works if cwd is elsewhere)."""
    cwd = Path(".").resolve()
    for p in [cwd] + list(cwd.parents):
        if (p / "PS3.ipynb").is_file():
            return p
    return cwd


ROOT = homework_dir()
MANIFEST_CSV = ROOT / "data" / "WebNet_Mar6" / "fullmix.csv"

device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
print("ROOT:", ROOT.resolve())
print("Manifest:", MANIFEST_CSV.resolve(), "exists:", MANIFEST_CSV.is_file())
print("Device:", device, "| torch:", torch.__version__)
```

```
ROOT: C:\Users\mpave\OneDrive - Massachusetts Institute of Technology\2.174 Mechanic
s and Materials via Machine Learning\PS3
Manifest: C:\Users\mpave\OneDrive - Massachusetts Institute of Technology\2.174 Mech
anics and Materials via Machine Learning\PS3\data\WebNet_Mar6\fullmix.csv exists: Tr
ue
Device: cpu | torch: 2.11.0+cpu
```

In [31]:

```python
# --- Build PyG Data list from manifest (same logic as WebNet_GNN.ipynb) ---


def graph_folder(rel_dir: str) -> Path:
    """Turn CSV dir entry into absolute path under ROOT/data/..."""
    rel = rel_dir.replace("\\", "/").lstrip("./")
    return (ROOT / "data" / rel).resolve()


def build_data_list(pro_type: str):
    """pro_type: 'Strength' or 'Toughness'."""
    if pro_type not in ("Strength", "Toughness"):
        raise ValueError("pro_type must be 'Strength' or 'Toughness'")

    graph_label_raw = pd.read_csv(MANIFEST_CSV)
    filelist = graph_label_raw["dir"]
    if pro_type == "Strength":
        graph_label_list = graph_label_raw["AverageStrength"]
    else:
```

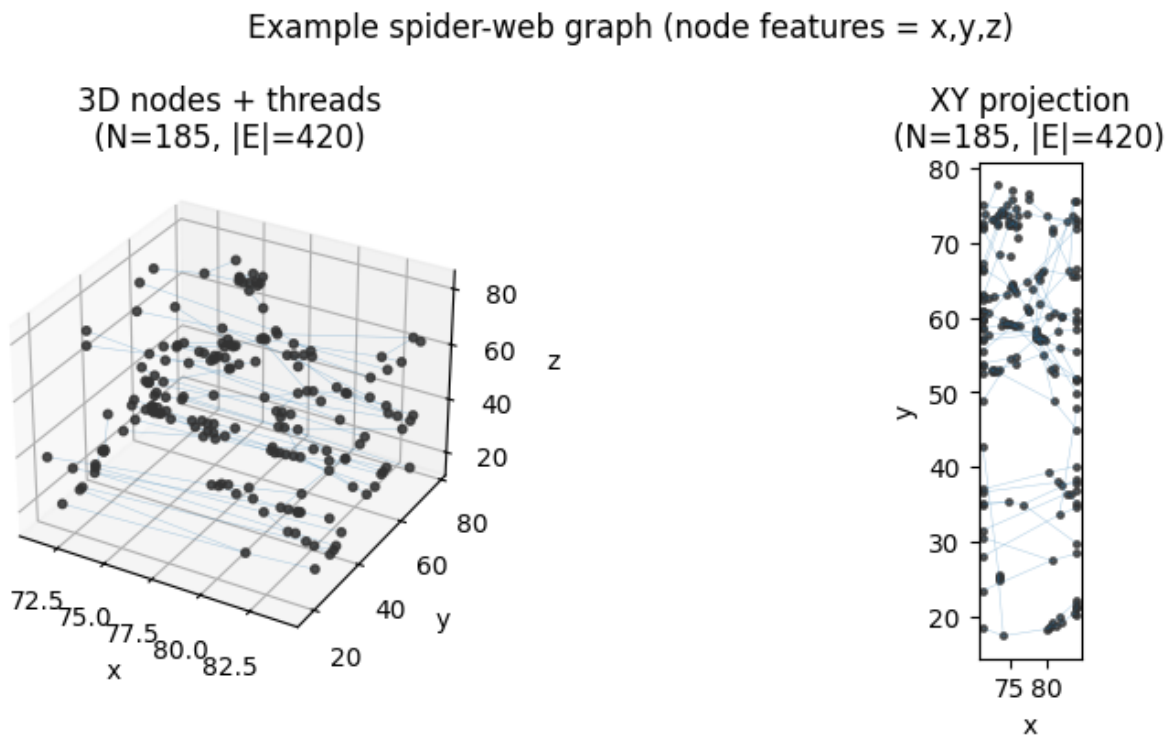
```
        graph_label_list = graph_label_raw["AverageToughness"]

    data_list = []
    for i in range(len(filelist)):
        folder = graph_folder(str(filelist.iloc[i]))
        try:
            node_raw = pd.read_csv(folder / "atomXYZ_final.csv")
            edge_raw = pd.read_csv(folder / "bond_final.csv")

            nx = np.array(node_raw["x"]).reshape(-1, 1)
            ny = np.array(node_raw["y"]).reshape(-1, 1)
            nz = np.array(node_raw["z"]).reshape(-1, 1)
            node_list = np.hstack((nx, ny, nz))

            n1 = np.array(edge_raw["node1"] - 1).reshape(-1, 1)
            n2 = np.array(edge_raw["node2"] - 1).reshape(-1, 1)
            a = np.hstack((n1, n2))
            b = np.hstack((n2, n1))
            ab = np.append(a, b, axis=0)
            edge_list = np.transpose(ab)

            graph_label = np.array(graph_label_list.iloc[i]).reshape(1, -1)

            x = torch.tensor(node_list, dtype=torch.float)
            edge_index = torch.tensor(edge_list, dtype=torch.long)
            y = torch.tensor(graph_label, dtype=torch.float)
            data_list.append(Data(x=x, edge_index=edge_index, y=y))
        except Exception as e:
            print(f"Skip index {i} ({folder}): {e}")

    return data_list
```

In [32]:

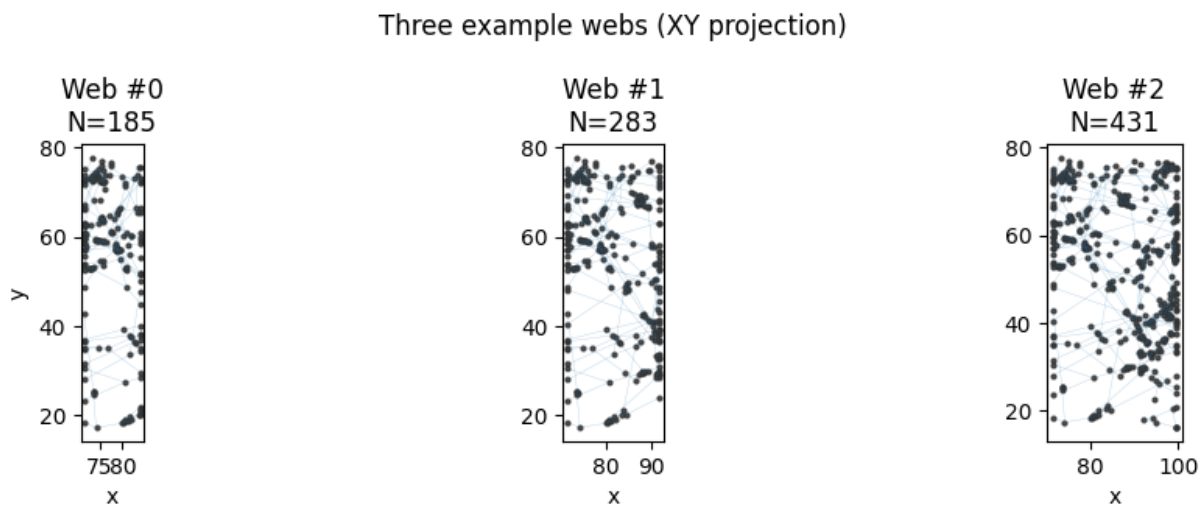
```
# --- Visualize a few WebNet graphs (3D + XY projection) ---
# Run after `homework_dir` / `build_data_list` are defined above.

import matplotlib.pyplot as plt

# Load one target only for plotting (geometry is the same for Strength vs Toughness
_viz_list = build_data_list("Strength")


def plot_single_web(ax3d, ax_xy, data, title_suffix=""):
    """Draw one graph: 3D nodes+edges and an XY top-down view."""
    pts = data.x.detach().cpu().numpy()
    ei = data.edge_index.detach().cpu().numpy()
    # Unique undirected edges for cleaner drawing (bond list is duplicated both way
    seen = set()
    edges = []
    for a, b in ei.T:
        u, v = int(a), int(b)
        if u == v:
            continue
        key = (u, v) if u < v else (v, u)
        if key in seen:
            continue
        seen.add(key)
```

```
        edges.append((u, v))

    # 3D
    ax3d.scatter(pts[:, 0], pts[:, 1], pts[:, 2], s=10, c="0.2", alpha=0.85, depths
    for u, v in edges:
        ax3d.plot(
            [pts[u, 0], pts[v, 0]],
            [pts[u, 1], pts[v, 1]],
            [pts[u, 2], pts[v, 2]],
            color="C0",
            lw=0.35,
            alpha=0.35,
        )
    ax3d.set_xlabel("x")
    ax3d.set_ylabel("y")
    ax3d.set_zlabel("z")
    ax3d.set_title("3D nodes + threads" + title_suffix)

    # XY (top-down)
    ax_xy.scatter(pts[:, 0], pts[:, 1], s=6, c="0.2", alpha=0.8)
    for u, v in edges:
        ax_xy.plot(
            [pts[u, 0], pts[v, 0]],
            [pts[u, 1], pts[v, 1]],
            color="C0",
            lw=0.35,
            alpha=0.35,
        )
    ax_xy.set_aspect("equal", adjustable="box")
    ax_xy.set_xlabel("x")
    ax_xy.set_ylabel("y")
    ax_xy.set_title("XY projection" + title_suffix)


# Figure 1 — first graph in detail
if _viz_list:
    d0 = _viz_list[0]
    fig = plt.figure(figsize=(10, 4.2))
    ax3d = fig.add_subplot(1, 2, 1, projection="3d")
    ax_xy = fig.add_subplot(1, 2, 2)
    plot_single_web(
        ax3d,
        ax_xy,
        d0,
        title_suffix=f"\n(N={d0.num_nodes}, |E|={d0.edge_index.shape[1]})",
    )
    fig.suptitle("Example spider-web graph (node features = x,y,z)", fontsize=12)
    plt.tight_layout()
    out1 = ROOT / "charts" / "webnet_gnn" / "sample_web_primary.png"
    out1.parent.mkdir(parents=True, exist_ok=True)
    fig.savefig(out1, dpi=150, bbox_inches="tight")
    plt.show()
    print("Saved:", out1.resolve())

# Figure 2 — three different webs, XY only (quick visual scan)
fig2, axes = plt.subplots(1, 3, figsize=(11, 3.4), sharex=False, sharey=False)
```

```
for j, ax in enumerate(axes.ravel()):
    if j >= len(_viz_list):
        ax.axis("off")
        continue
    dj = _viz_list[j]
    pts = dj.x.detach().cpu().numpy()
    ei = dj.edge_index.detach().cpu().numpy()
    seen = set()
    edges = []
    for a, b in ei.T:
        u, v = int(a), int(b)
        if u == v:
            continue
        key = (u, v) if u < v else (v, u)
        if key in seen:
            continue
        seen.add(key)
        edges.append((u, v))
    ax.scatter(pts[:, 0], pts[:, 1], s=4, c="0.2", alpha=0.85)
    for u, v in edges:
        ax.plot(
            [pts[u, 0], pts[v, 0]],
            [pts[u, 1], pts[v, 1]],
            color="C0",
            lw=0.25,
            alpha=0.35,
        )
    ax.set_aspect("equal", adjustable="box")
    ax.set_title(f"Web #{j}\nN={dj.num_nodes}")
    ax.set_xlabel("x")
    if j == 0:
        ax.set_ylabel("y")
fig2.suptitle("Three example webs (XY projection)", fontsize=12)
plt.tight_layout()
out2 = ROOT / "charts" / "webnet_gnn" / "sample_web_triptych.png"
out2.parent.mkdir(parents=True, exist_ok=True)
fig2.savefig(out2, dpi=150, bbox_inches="tight")
plt.show()
print("Saved:", out2.resolve())
```




```
Saved: C:\Users\mpave\OneDrive - Massachusetts Institute of Technology\2.174 Mechani
cs and Materials via Machine Learning\PS3\charts\webnet_gnn\sample_web_primary.png
```




```
Saved: C:\Users\mpave\OneDrive - Massachusetts Institute of Technology\2.174 Mechani
cs and Materials via Machine Learning\PS3\charts\webnet_gnn\sample_web_triptych.png
```

In [33]:
```python
# --- GCN regression model (graph-level mean pool + linear head) ---


class GCN(torch.nn.Module):
    def __init__(self, in_channels: int, hidden_channels: int, out_channels: int =
        super().__init__()
        torch.manual_seed(42)
        self.conv1 = GCNConv(in_channels, hidden_channels)
        self.conv2 = GCNConv(hidden_channels, hidden_channels)
        self.conv3 = GCNConv(hidden_channels, hidden_channels)
        self.lin = Linear(hidden_channels, out_channels)

    def forward(self, x, edge_index, batch):
```

```
        x = x.to(self.lin.weight.device)
        edge_index = edge_index.to(self.lin.weight.device)
        batch = batch.to(self.lin.weight.device)
        x = self.conv1(x, edge_index).relu()
        x = self.conv2(x, edge_index).relu()
        x = self.conv3(x, edge_index)
        x = global_mean_pool(x, batch)
        x = F.dropout(x, p=0.2, training=self.training)
        return self.lin(x)
```

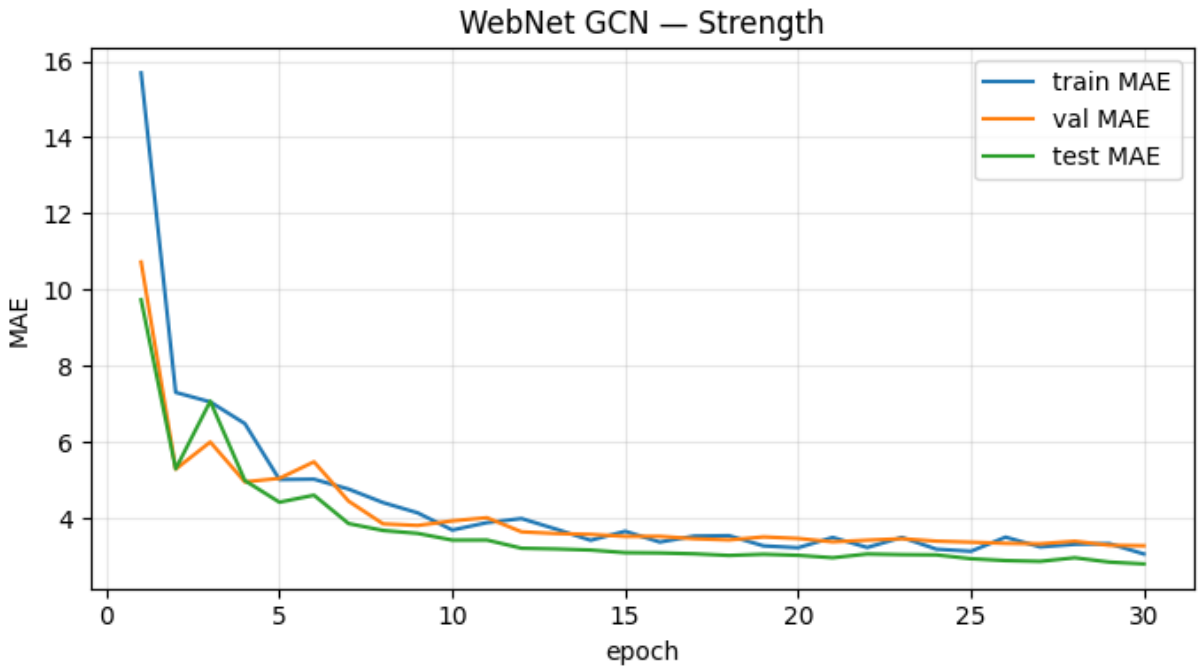
In [34]:

```
# --- Train / validate / test + save curves (per target) ---


def run_training(
    pro_type: str,
    *,
    hidden_channels: int = 32,
    batch_size: int = 32,
    epochs: int = 30,
    lr: float = 0.002,
    random_state: int = 47,
):
    data_list = build_data_list(pro_type)
    if not data_list:
        raise RuntimeError("No graphs loaded; check manifest paths under data/WebNe

    in_ch = int(data_list[0].num_node_features)

    Train_data, Test_data = train_test_split(data_list, test_size=0.4, random_state
    Test_data, Val_data = train_test_split(Test_data, test_size=0.5, random_state=r

    train_loader = DataLoader(Train_data, batch_size=batch_size, shuffle=True)
    val_loader = DataLoader(Val_data, batch_size=batch_size, shuffle=True)
    test_loader = DataLoader(Test_data, batch_size=batch_size, shuffle=False)

    print(
        f"[{pro_type}] train={len(Train_data)} val={len(Val_data)} test={len(Test_d
        f"| in_features={in_ch}"
    )

    out_root = ROOT / "charts" / "webnet_gnn" / "GCN" / pro_type
    best_dir = out_root / "best_model"
    plot_dir = out_root / "plots"
    for d in (best_dir, plot_dir):
        d.mkdir(parents=True, exist_ok=True)

    model = GCN(in_ch, hidden_channels, 1).to(device)
    optimizer = torch.optim.Adam(model.parameters(), lr=lr)
    scheduler = ReduceLROnPlateau(optimizer, mode="min", factor=0.5, patience=12, m
    mae = torch.nn.L1Loss()

    def train_epoch():
        model.train()
        total, count = 0.0, 0
        for data in train_loader:
            data = data.to(device)
```

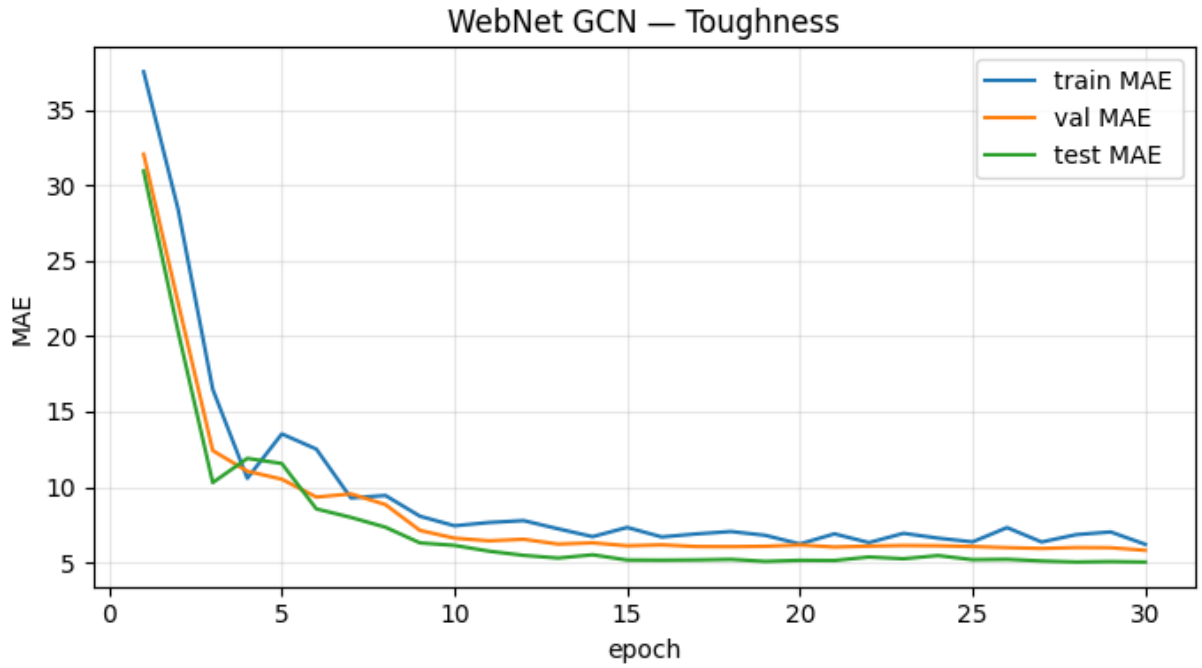
```
        optimizer.zero_grad()
        out = model(data.x, data.edge_index, data.batch)
        loss = mae(out.squeeze(), data.y.squeeze())
        loss.backward()
        total += float(loss.item()) * data.num_graphs
        count += data.num_graphs
        optimizer.step()
    return total / max(count, 1)

@torch.no_grad()
def eval_mae(loader):
    model.eval()
    total, count = 0.0, 0
    for data in loader:
        data = data.to(device)
        out = model(data.x, data.edge_index, data.batch)
        loss = mae(out.squeeze(), data.y.squeeze())
        total += float(loss.item()) * data.num_graphs
        count += data.num_graphs
    return total / max(count, 1)

train_maes, val_maes, test_maes = [], [], []
best_val = float("inf")

for epoch in tqdm(range(1, epochs + 1), desc=f"GCN / {pro_type}"):
    tr = train_epoch()
    va = eval_mae(val_loader)
    te = eval_mae(test_loader)
    train_maes.append(tr)
    val_maes.append(va)
    test_maes.append(te)
    scheduler.step(va)
    if va < best_val:
        best_val = va
        torch.save(model.state_dict(), best_dir / "model.pt")

import matplotlib.pyplot as plt

xs = np.arange(1, epochs + 1)
plt.figure(figsize=(7, 4))
plt.plot(xs, train_maes, label="train MAE")
plt.plot(xs, val_maes, label="val MAE")
plt.plot(xs, test_maes, label="test MAE")
plt.xlabel("epoch")
plt.ylabel("MAE")
plt.title(f"WebNet GCN — {pro_type}")
plt.legend()
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.savefig(plot_dir / "loss_curves.png", dpi=150)
plt.show()

print(f"[{pro_type}] best val MAE ≈ {best_val:.4f} | weights → {best_dir / 'mod
print(f"[{pro_type}] plot → {plot_dir / 'loss_curves.png'}")
```

In [35]:
```
# --- Part (a): train two separate GCN models (strength, then toughness) ---
# Lower `epochs` while debugging; increase for final submission.

run_training("Strength", epochs=30)
run_training("Toughness", epochs=30)
```

```
[Strength] train=120 val=40 test=40 | in_features=3
GCN / Strength: 100%|██████████| 30/30 [00:21<00:00,  1.36it/s]
```




```
[Strength] best val MAE ≈ 3.2757 | weights → C:\Users\mpave\OneDrive - Massachusetts
Institute of Technology\2.174 Mechanics and Materials via Machine Learning\PS3\chart
s\webnet_gnn\GCN\Strength\best_model\model.pt
[Strength] plot → C:\Users\mpave\OneDrive - Massachusetts Institute of Technology
\2.174 Mechanics and Materials via Machine Learning\PS3\charts\webnet_gnn\GCN\Streng
th\plots\loss_curves.png
[Toughness] train=120 val=40 test=40 | in_features=3
GCN / Toughness: 100%|██████████| 30/30 [00:21<00:00,  1.37it/s]
```




```
[Toughness] best val MAE ≈ 5.8137 | weights → C:\Users\mpave\OneDrive - Massachusett
s Institute of Technology\2.174 Mechanics and Materials via Machine Learning\PS3\cha
rts\webnet_gnn\GCN\Toughness\best_model\model.pt
[Toughness] plot → C:\Users\mpave\OneDrive - Massachusetts Institute of Technology
\2.174 Mechanics and Materials via Machine Learning\PS3\charts\webnet_gnn\GCN\Toughn
ess\plots\loss_curves.png
```

(B)

I trained two GCNs (one for strength and one for toughness) and plotted MAE vs epoch for train, validation, and test. Both errors drop quickly at first then levels off after about 15 epochs. Train, val, and test stay close with no rising val/test while train keeps improving showing no strong overfitting.

Strength ends near MAE = 3 on all splits. Toughness was slightly different, final MAEs are about 6 (train), 6 (val), and 5 (test) which is larger than strength but the same qualitative pattern.

(C)

In [37]:
```
import random


def _data_clone(d: Data) -> Data:
    return Data(x=d.x.clone(), edge_index=d.edge_index.clone(), y=d.y.clone())


def _undirected_unique_pairs(edge_index: torch.Tensor):
    ei = edge_index.cpu().numpy()
    seen = set()
    pairs = []
    for j in range(ei.shape[1]):
```

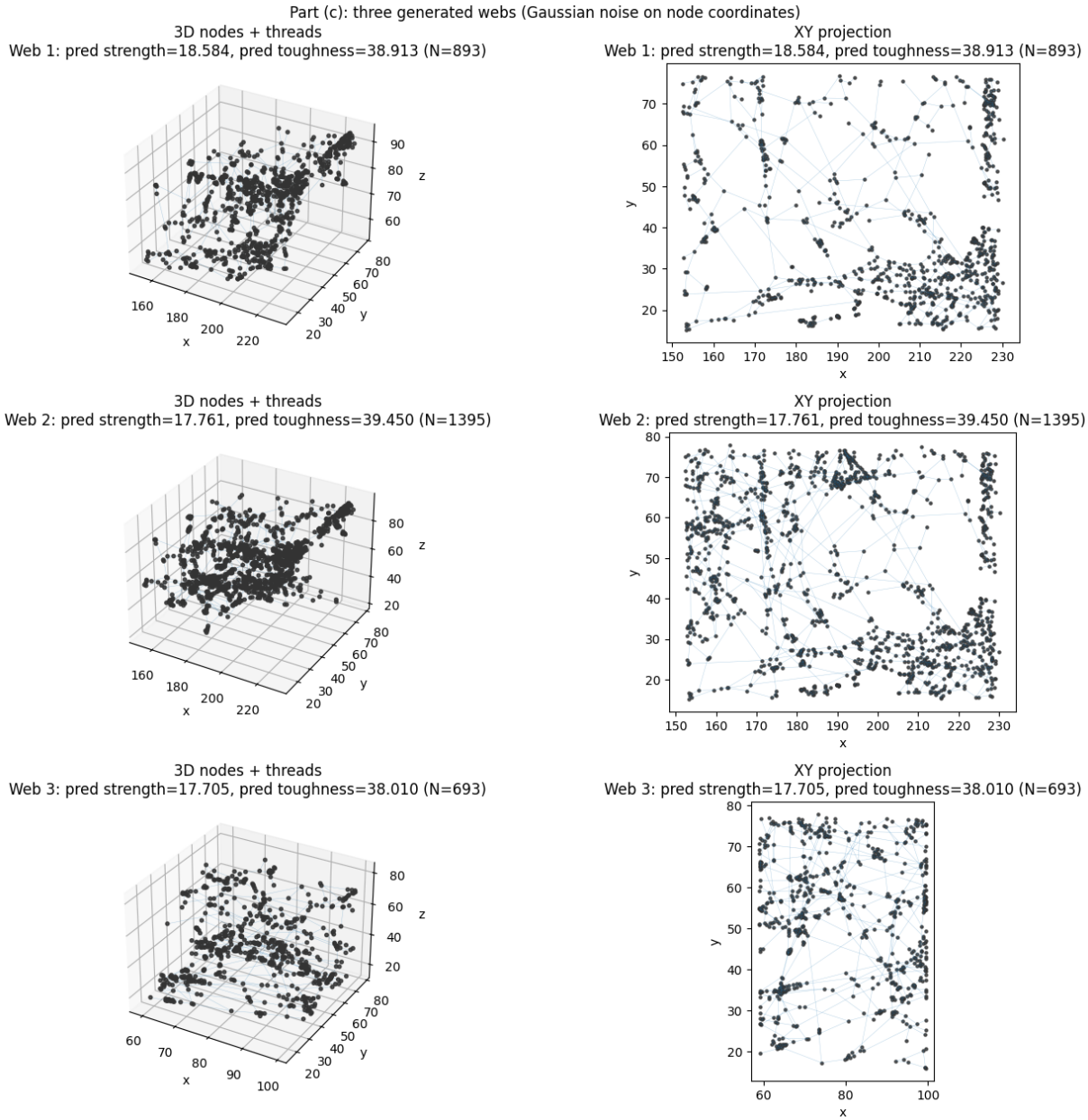
```
        a, b = int(ei[0, j]), int(ei[1, j])
        if a == b:
            continue
        key = (a, b) if a < b else (b, a)
        if key in seen:
            continue
        seen.add(key)
        pairs.append(key)
    return pairs


def augment_node_noise(d: Data, sigma: float) -> Data:
    out = _data_clone(d)
    out.x = out.x + torch.randn_like(out.x) * sigma
    return out


def augment_edge_dropout(d: Data, drop_frac: float, rng: random.Random) -> Data:
    """Drop a fraction of undirected bonds; keep graph connected only in expectatio
    out = _data_clone(d)
    pairs = _undirected_unique_pairs(out.edge_index)
    if not pairs:
        return out
    kept = [e for e in pairs if rng.random() > drop_frac]
    if not kept:
        return out
    a = np.array([e[0] for e in kept], dtype=np.int64)
    b = np.array([e[1] for e in kept], dtype=np.int64)
    ei = np.stack([np.concatenate([a, b]), np.concatenate([b, a])], axis=0)
    out.edge_index = torch.tensor(ei, dtype=torch.long)
    return out


def graph_stats(d: Data):
    pairs = _undirected_unique_pairs(d.edge_index)
    return {"nodes": int(d.num_nodes), "edges_ud": len(pairs)}


@torch.no_grad()
def predict_list(model: GCN, graphs: list) -> np.ndarray:
    loader = DataLoader(graphs, batch_size=32, shuffle=False)
    parts = []
    for data in loader:
        data = data.to(device)
        out = model(data.x, data.edge_index, data.batch)
        parts.append(out.view(-1).cpu().numpy())
    return np.concatenate(parts, axis=0)


def load_trained_gcn(pro_type: str, in_ch: int, hidden_channels: int = 32) -> GCN:
    wpath = ROOT / "charts" / "webnet_gnn" / "GCN" / pro_type / "best_model" / "mod
    if not wpath.is_file():
        raise FileNotFoundError(f"Missing weights (train Part (a) first): {wpath}")
    m = GCN(in_ch, hidden_channels, 1).to(device)
    m.load_state_dict(torch.load(wpath, map_location=device))
    m.eval()
```

```
    return m


def summarize(name: str, y_true: np.ndarray, pred_orig: np.ndarray, pred_aug: np.nd
    mae_orig = float(np.mean(np.abs(pred_orig - y_true)))
    mae_aug = float(np.mean(np.abs(pred_aug - y_true)))
    mean_d = float(np.mean(np.abs(pred_aug - pred_orig)))
    print(f"{name}: MAE(orig)={mae_orig:.3f}  MAE(aug)={mae_aug:.3f}  mean|Δpred|={


# --- run a fixed subsample for speed ---
rng_py = random.Random(0)
torch.manual_seed(0)

strength_graphs = build_data_list("Strength")
toughness_graphs = build_data_list("Toughness")
n = min(len(strength_graphs), len(toughness_graphs))
NUM_SEED_WEBS = 3  # exactly this many new (augmented) webs below
ix = list(range(n))
rng_py.shuffle(ix)
ix = ix[: min(NUM_SEED_WEBS, n)]

orig_s = [strength_graphs[i] for i in ix]
orig_t = [toughness_graphs[i] for i in ix]

SIGMA = 0.05  # noise on coords; reduce if predictions blow up

noise_s = [augment_node_noise(g, SIGMA) for g in orig_s]

# Graph statistics (3 seeds → 3 new webs via node noise only)
for label, seq in [("orig", orig_s), ("+node noise (3 new webs)", noise_s)]:
    nodes = [graph_stats(d)["nodes"] for d in seq]
    edges = [graph_stats(d)["edges_ud"] for d in seq]
    print(f"[stats {label}]  nodes: mean={np.mean(nodes):.1f} std={np.std(nodes):.1
          f"edges (undir): mean={np.mean(edges):.1f} std={np.std(edges):.1f}")

in_ch = int(orig_s[0].num_node_features)
model_s = load_trained_gcn("Strength", in_ch)
model_t = load_trained_gcn("Toughness", in_ch)

y_s = np.array([float(g.y.ravel()[0]) for g in orig_s], dtype=np.float64)
y_t = np.array([float(g.y.ravel()[0]) for g in orig_t], dtype=np.float64)

# Strength model
p0 = predict_list(model_s, orig_s)
pred_strength_aug = predict_list(model_s, noise_s)
summarize("Strength | node noise", y_s, p0, pred_strength_aug)

# Toughness model (same augmented geometries; labels from toughness manifest)
p0t = predict_list(model_t, orig_t)
pred_toughness_aug = predict_list(model_t, noise_s)
summarize("Toughness | node noise", y_t, p0t, pred_toughness_aug)

# --- Predicted strength & toughness for each *generated* web (trained GCNs) ---
dfc = pd.DataFrame(
    {
```

```
        "web": np.arange(1, len(noise_s) + 1, dtype=int),
        "manifest_idx": ix,
        "pred_strength": pred_strength_aug,
        "pred_toughness": pred_toughness_aug,
        "measured_strength_seed": y_s,
        "measured_toughness_seed": y_t,
    }
)
print("\n=== Part (c): three generated webs - GCN predictions ===")
print("(Geometry is noisy; labels in the last two columns are the *seed* web measur
print(dfc.to_string(index=False))

# --- Plot geometry of the three generated webs (3D + XY per row) ---
import matplotlib.pyplot as plt


def _plot_web_panels(ax3d, ax_xy, data, title_3d: str, title_xy: str):
    pts = data.x.detach().cpu().numpy()
    ei = data.edge_index.detach().cpu().numpy()
    seen = set()
    edges = []
    for a, b in ei.T:
        u, v = int(a), int(b)
        if u == v:
            continue
        key = (u, v) if u < v else (v, u)
        if key in seen:
            continue
        seen.add(key)
        edges.append((u, v))

    ax3d.scatter(pts[:, 0], pts[:, 1], pts[:, 2], s=8, c="0.2", alpha=0.85, depthsh
    for u, v in edges:
        ax3d.plot(
            [pts[u, 0], pts[v, 0]],
            [pts[u, 1], pts[v, 1]],
            [pts[u, 2], pts[v, 2]],
            color="C0",
            lw=0.3,
            alpha=0.35,
        )
    ax3d.set_xlabel("x")
    ax3d.set_ylabel("y")
    ax3d.set_zlabel("z")
    ax3d.set_title(title_3d)

    ax_xy.scatter(pts[:, 0], pts[:, 1], s=5, c="0.2", alpha=0.8)
    for u, v in edges:
        ax_xy.plot(
            [pts[u, 0], pts[v, 0]],
            [pts[u, 1], pts[v, 1]],
            color="C0",
            lw=0.3,
            alpha=0.35,
        )
    ax_xy.set_aspect("equal", adjustable="box")
```

```
    ax_xy.set_xlabel("x")
    ax_xy.set_ylabel("y")
    ax_xy.set_title(title_xy)


fig = plt.figure(figsize=(15, 4.3 * len(noise_s)))
for k, d in enumerate(noise_s):
    tline = (
        f"Web {k + 1}: pred strength={pred_strength_aug[k]:.3f}, "
        f"pred toughness={pred_toughness_aug[k]:.3f} (N={d.num_nodes})"
    )
    ax3d = fig.add_subplot(len(noise_s), 2, 2 * k + 1, projection="3d")
    ax_xy = fig.add_subplot(len(noise_s), 2, 2 * k + 2)
    _plot_web_panels(
        ax3d,
        ax_xy,
        d,
        title_3d="3D nodes + threads\n" + tline,
        title_xy="XY projection\n" + tline,
    )

fig.suptitle("Part (c): three generated webs (Gaussian noise on node coordinates)",
fig.tight_layout()
_out = ROOT / "charts" / "webnet_gnn" / "part_c_three_generated_webs.png"
_out.parent.mkdir(parents=True, exist_ok=True)
fig.savefig(_out, dpi=150, bbox_inches="tight")
plt.show()
print("Saved:", _out.resolve())
```

```
[stats orig]  nodes: mean=993.7 std=295.3   edges (undir): mean=1367.0 std=440.0
[stats +node noise (3 new webs)]  nodes: mean=993.7 std=295.3   edges (undir): mean=
1367.0 std=440.0
Strength | node noise: MAE(orig)=3.175  MAE(aug)=3.175  mean|Δpred|=0.000
Toughness | node noise: MAE(orig)=6.452  MAE(aug)=6.452  mean|Δpred|=0.000

=== Part (c): three generated webs - GCN predictions ===
(Geometry is noisy; labels in the last two columns are the *seed* web measurements.)

 web  manifest_idx  pred_strength  pred_toughness  measured_strength_seed  measured_
toughness_seed
   1           131      18.584370       38.912670               18.526293
36.329350
   2           181      17.760571       39.449612               14.649590
32.682945
   3            22      17.705307       38.009750               24.060429
48.015343
```




```
Saved: C:\Users\mpave\OneDrive - Massachusetts Institute of Technology\2.174 Mechani
cs and Materials via Machine Learning\PS3\charts\webnet_gnn\part_c_three_generated_w
ebs.png
```

(D)

For manufacturing spider-web-like lattice geometries one option is two-photon lithographysince it can fabricate true 3D microstructures with very high spatial resolution and complex free-standing architectures. This makes it well-suited for reproducing thin thread networks and junctions. For larger millimeter-scale structures lower-resolution additive methods (e.g SLA) could approximate the geometry, but TPL is the best match when fine features and 3D connectivity matter most (Harinarayana & Shin, *Optics & Laser Technology*, 2021).

Problem 3

(A)

In [1]:
```
import math
import random
import numpy as np
import torch
import torch.nn as nn
import torch.nn.functional as F
import matplotlib.pyplot as plt

# ----------------------------
# Reproducibility + device
# ----------------------------
random.seed(42)
np.random.seed(42)
torch.manual_seed(42)

if torch.cuda.is_available():
    device = torch.device("cuda")
elif torch.backends.mps.is_available():
    device = torch.device("mps")
else:
    device = torch.device("cpu")

print("Using device:", device)

# ----------------------------
# UniRef50 data pipeline
# ----------------------------
try:
    from datasets import load_dataset
except ImportError:
    import subprocess
    import sys
    subprocess.check_call([sys.executable, "-m", "pip", "install", "datasets"])
    from datasets import load_dataset

N_TRAIN = 20_000
HF_DATASET = "lamm-mit/UniRef50_512_500K"

print(f"Loading {HF_DATASET} with split train[:{N_TRAIN}] ...")
raw = load_dataset(HF_DATASET, split=f"train[:{N_TRAIN}]")

corpus = [row["Sequence"].strip().upper() for row in raw]
corpus = [s for s in corpus if s]
print(f"Number of training sequences: {len(corpus)}")
print("Example sequences:")
for s in corpus[:3]:
    print(" -", s[:80] + ("..." if len(s) > 80 else ""))

# ----------------------------
```

```python
# Build vocabulary
# ----------------------------
special_tokens = ["<pad>", "<bos>", "<eos>", "<unk>"]

aas = set()
for line in corpus:
    for ch in line:
        if ch.isalpha():
            aas.add(ch)

vocab = special_tokens + sorted(aas)
stoi = {s: i for i, s in enumerate(vocab)}
itos = {i: s for s, i in stoi.items()}

PAD = stoi["<pad>"]
BOS = stoi["<bos>"]
EOS = stoi["<eos>"]
UNK = stoi["<unk>"]
vocab_size = len(vocab)

print("\nVocab size:", vocab_size)
print("Vocabulary:", vocab)

def tokenize(text, add_bos=True, add_eos=True):
    toks = [c for c in text.strip().upper() if c.isalpha()]
    ids = []
    if add_bos:
        ids.append(BOS)
    ids.extend([stoi.get(tok, UNK) for tok in toks])
    if add_eos:
        ids.append(EOS)
    return ids

def detokenize(token_ids, remove_special=True):
    toks = []
    for idx in token_ids:
        tok = itos.get(int(idx), "<unk>")
        if remove_special and tok in {"<pad>", "<bos>", "<eos>"}:
            continue
        toks.append(tok)
    return "".join(toks)

# ----------------------------
# Sequence prep
# ----------------------------
tokenized_corpus = [tokenize(line) for line in corpus]
max_len = max(len(x) for x in tokenized_corpus)
print("Max sequence length:", max_len)

def sample_batch(batch_size=16):
    batch = random.choices(tokenized_corpus, k=batch_size)
    x_batch, y_batch = [], []
    for ids in batch:
        x = ids[:-1]
        y = ids[1:]
        pad_x = x + [PAD] * ((max_len - 1) - len(x))
```

```
        pad_y = y + [PAD] * ((max_len - 1) - len(y))
        x_batch.append(pad_x)
        y_batch.append(pad_y)
    return (
        torch.tensor(x_batch, dtype=torch.long, device=device),
        torch.tensor(y_batch, dtype=torch.long, device=device),
    )
```

```
Using device: cpu
c:\Users\mpave\OneDrive - Massachusetts Institute of Technology\2.174 Mechanics and
Materials via Machine Learning\PS3\.venv\Lib\site-packages\tqdm\auto.py:21: TqdmWarn
ing: IProgress not found. Please update jupyter and ipywidgets. See https://ipywidge
ts.readthedocs.io/en/stable/user_install.html
  from .autonotebook import tqdm as notebook_tqdm
Loading lamm-mit/UniRef50_512_500K with split train[:20000] ...
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOK
EN to enable higher rate limits and faster downloads.
c:\Users\mpave\OneDrive - Massachusetts Institute of Technology\2.174 Mechanics and
Materials via Machine Learning\PS3\.venv\Lib\site-packages\huggingface_hub\file_down
load.py:138: UserWarning: `huggingface_hub` cache-system uses symlinks by default to
efficiently store duplicated files but your machine does not support them in C:\User
s\mpave\.cache\huggingface\hub\datasets--lamm-mit--UniRef50_512_500K. Caching files
will still work but in a degraded version that might require more space on your disk
. This warning can be disabled by setting the `HF_HUB_DISABLE_SYMLINKS_WARNING` envi
ronment variable. For more details, see https://huggingface.co/docs/huggingface_hub/
how-to-cache#limitations.
To support symlinks on Windows, you either need to activate Developer Mode or to run
Python as an administrator. In order to activate developer mode, see this article: h
ttps://docs.microsoft.com/en-us/windows/apps/get-started/enable-your-device-for-deve
lopment
  warnings.warn(message)
Generating train split: 100%|██████████| 500000/500000 [00:00<00:00, 2188787.79 exam
ples/s]
Number of training sequences: 20000
Example sequences:
 - MGYIHTALKSAGFHHVIQVDTPALGLDSEGLRKLLADFEPDLVGVSTTTPGLPGAIEACEAAKSTGAKVILGGPHTEVY
A...
 - MAKSNQKSDSKAFVLALNHEMIYADKILKFGIFTL
 - MRCAKHSLYGFAVPFFSQKGHRLFYLYIQQ

Vocab size: 26
Vocabulary: ['<pad>', '<bos>', '<eos>', '<unk>', 'A', 'C', 'D', 'E', 'F', 'G', 'H',
'I', 'K', 'L', 'M', 'N', 'P', 'Q', 'R', 'S', 'T', 'U', 'V', 'W', 'X', 'Y']
Max sequence length: 514
```

In [2]:

```
# ----------------------------
# Positional encoding
# ----------------------------
class PositionalEncoding(nn.Module):
    def __init__(self, d_model, max_len=256):
        super().__init__()
        pe = torch.zeros(max_len, d_model)
        pos = torch.arange(0, max_len, dtype=torch.float32).unsqueeze(1)
        div = torch.exp(torch.arange(0, d_model, 2, dtype=torch.float32) * (-math.l
        pe[:, 0::2] = torch.sin(pos * div)
        pe[:, 1::2] = torch.cos(pos * div)
```

```python
        self.register_buffer("pe", pe.unsqueeze(0))  # (1, T, C)

    def forward(self, x):
        return x + self.pe[:, :x.size(1)]

# ----------------------------
# Multi-head self-attention
# ----------------------------
class MultiHeadSelfAttention(nn.Module):
    def __init__(self, d_model, n_heads, dropout=0.1):
        super().__init__()
        assert d_model % n_heads == 0, "d_model must be divisible by n_heads"
        self.n_heads = n_heads
        self.head_dim = d_model // n_heads

        self.q_proj = nn.Linear(d_model, d_model)
        self.k_proj = nn.Linear(d_model, d_model)
        self.v_proj = nn.Linear(d_model, d_model)
        self.out_proj = nn.Linear(d_model, d_model)
        self.dropout = nn.Dropout(dropout)

        self.last_attn = None  # store for visualization

    def forward(self, x, attn_mask=None):
        B, T, C = x.shape

        q = self.q_proj(x).view(B, T, self.n_heads, self.head_dim).transpose(1, 2)
        k = self.k_proj(x).view(B, T, self.n_heads, self.head_dim).transpose(1, 2)
        v = self.v_proj(x).view(B, T, self.n_heads, self.head_dim).transpose(1, 2)

        scores = torch.matmul(q, k.transpose(-2, -1)) / math.sqrt(self.head_dim)  #

        if attn_mask is not None:
            scores = scores.masked_fill(~attn_mask, float("-inf"))

        attn = F.softmax(scores, dim=-1)
        attn = self.dropout(attn)
        self.last_attn = attn.detach()

        context = torch.matmul(attn, v)  # (B,H,T,D)
        context = context.transpose(1, 2).contiguous().view(B, T, C)
        return self.out_proj(context)

# ----------------------------
# Feed-forward block
# ----------------------------
class FeedForward(nn.Module):
    def __init__(self, d_model, d_ff, dropout=0.1):
        super().__init__()
        self.net = nn.Sequential(
            nn.Linear(d_model, d_ff),
            nn.GELU(),
            nn.Linear(d_ff, d_model),
            nn.Dropout(dropout),
        )
```

```
    def forward(self, x):
        return self.net(x)

# ----------------------------
# Transformer block
# ----------------------------
class TransformerBlock(nn.Module):
    def __init__(self, d_model, n_heads, d_ff, dropout=0.1):
        super().__init__()
        self.ln1 = nn.LayerNorm(d_model)
        self.attn = MultiHeadSelfAttention(d_model, n_heads, dropout=dropout)
        self.ln2 = nn.LayerNorm(d_model)
        self.ff = FeedForward(d_model, d_ff, dropout=dropout)

    def forward(self, x, attn_mask=None):
        x = x + self.attn(self.ln1(x), attn_mask=attn_mask)
        x = x + self.ff(self.ln2(x))
        return x

# ----------------------------
# Minimal GPT
# ----------------------------
class MiniGPT(nn.Module):
    def __init__(self, vocab_size, d_model=96, n_heads=4, d_ff=256, n_layers=3, max
        super().__init__()
        self.token_emb = nn.Embedding(vocab_size, d_model)
        self.pos_enc = PositionalEncoding(d_model, max_len=max_len)
        self.blocks = nn.ModuleList([
            TransformerBlock(d_model, n_heads, d_ff, dropout=dropout)
            for _ in range(n_layers)
        ])
        self.ln_f = nn.LayerNorm(d_model)
        self.head = nn.Linear(d_model, vocab_size)

    def make_causal_pad_mask(self, x):
        # x: (B,T)
        B, T = x.shape
        causal = torch.tril(torch.ones(T, T, dtype=torch.bool, device=x.device))  #
        causal = causal.unsqueeze(0).unsqueeze(0)  # (1,1,T,T)

        # keys that are real tokens (not PAD)
        key_padding = (x != PAD).unsqueeze(1).unsqueeze(2)  # (B,1,1,T)

        # combine: can only attend to past AND non-pad keys
        mask = causal & key_padding  # (B,1,T,T) by broadcasting
        return mask

    def forward(self, x):
        attn_mask = self.make_causal_pad_mask(x)
        x = self.token_emb(x)
        x = self.pos_enc(x)
        for block in self.blocks:
            x = block(x, attn_mask=attn_mask)
        x = self.ln_f(x)
        logits = self.head(x)
        return logits
```

```
model = MiniGPT(
    vocab_size=vocab_size,
    d_model=96,
    n_heads=4,
    d_ff=256,
    n_layers=3,
    max_len=max_len,
    dropout=0.1,
).to(device)

optimizer = torch.optim.AdamW(model.parameters(), lr=2e-3, weight_decay=1e-4)
loss_fn = nn.CrossEntropyLoss(ignore_index=PAD)

# ----------------------------
# Training
# ----------------------------
steps = 800
batch_size = 32
loss_history = []

model.train()
for step in range(steps):
    xb, yb = sample_batch(batch_size=batch_size)
    logits = model(xb)  # (B,T,V)
    loss = loss_fn(logits.reshape(-1, vocab_size), yb.reshape(-1))

    optimizer.zero_grad()
    loss.backward()
    optimizer.step()

    loss_history.append(loss.item())
    if step % 100 == 0 or step == steps - 1:
        print(f"Step {step:4d} | Loss {loss.item():.4f}")
```

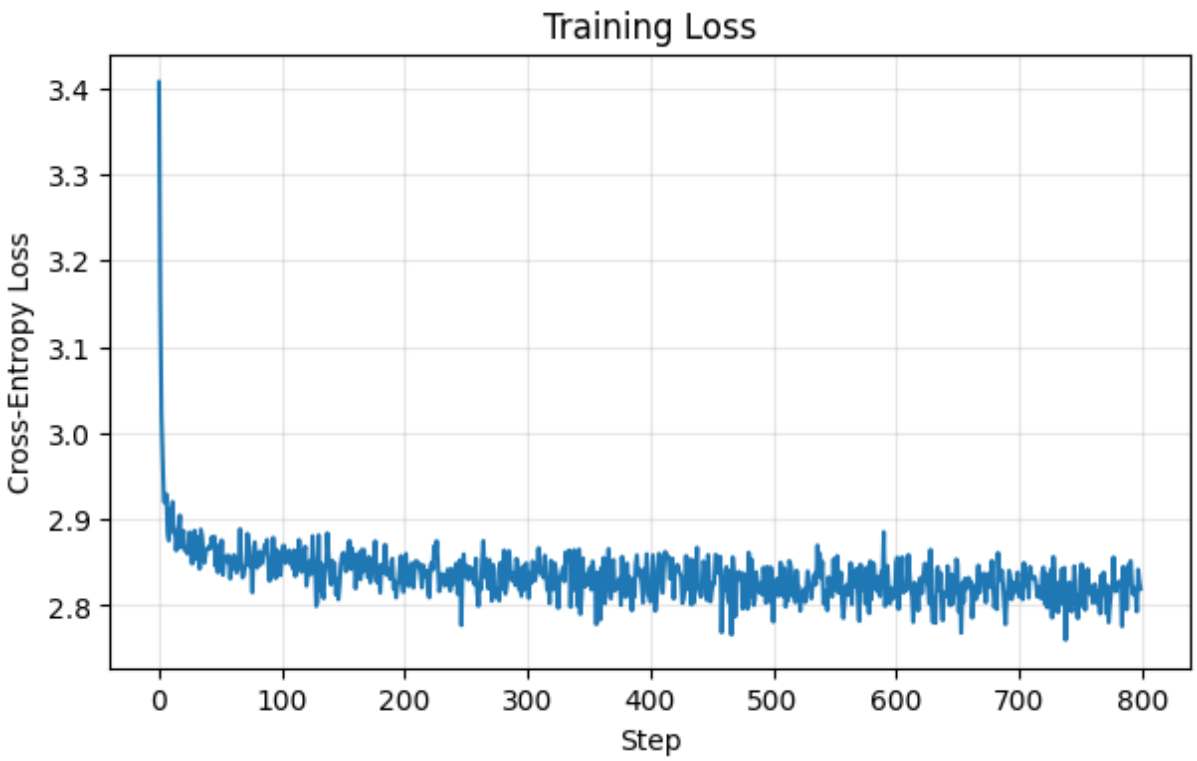
```
Step    0 | Loss 3.4070
Step  100 | Loss 2.8605
Step  200 | Loss 2.8460
Step  300 | Loss 2.8331
Step  400 | Loss 2.8045
Step  500 | Loss 2.7812
Step  600 | Loss 2.8500
Step  700 | Loss 2.8104
Step  799 | Loss 2.8193
```

In [3]:

```
# ----------------------------
# Generation
# ----------------------------
@torch.no_grad()
def generate(prompt, max_new_tokens=12, temperature=1.0, top_k=5):
    model.eval()

    ids = tokenize(prompt, add_bos=True, add_eos=False)
    x = torch.tensor([ids], dtype=torch.long, device=device)

    for _ in range(max_new_tokens):
```

```
        logits = model(x)
        next_logits = logits[:, -1, :] / temperature

        if top_k is not None:
            vals, _ = torch.topk(next_logits, k=min(top_k, next_logits.shape[-1]))
            kth = vals[:, -1].unsqueeze(-1)
            next_logits = torch.where(next_logits < kth, torch.full_like(next_logit

        probs = F.softmax(next_logits, dim=-1)
        next_token = torch.multinomial(probs, num_samples=1)

        if next_token.item() == EOS:
            break

        x = torch.cat([x, next_token], dim=1)

    return detokenize(x[0].tolist())

# ----------------------------
# Visualization helpers
# ----------------------------
def plot_loss(loss_history):
    plt.figure(figsize=(7, 4))
    plt.plot(loss_history)
    plt.title("Training Loss")
    plt.xlabel("Step")
    plt.ylabel("Cross-Entropy Loss")
    plt.grid(alpha=0.3)
    plt.show()

@torch.no_grad()
def plot_next_token_distribution(prompt, top_n=10, temperature=1.0):
    model.eval()
    ids = tokenize(prompt, add_bos=True, add_eos=False)
    x = torch.tensor([ids], dtype=torch.long, device=device)
    logits = model(x)
    next_logits = logits[0, -1] / temperature
    probs = F.softmax(next_logits, dim=0).detach().cpu().numpy()

    top_idx = np.argsort(probs)[::-1][:top_n]
    top_tokens = [itos[i] for i in top_idx]
    top_probs = probs[top_idx]

    plt.figure(figsize=(8, 4))
    plt.bar(range(len(top_tokens)), top_probs)
    plt.xticks(range(len(top_tokens)), top_tokens, rotation=45)
    plt.title(f"Next-token probabilities after: '{prompt}'")
    plt.ylabel("Probability")
    plt.xlabel("Candidate token")
    plt.grid(axis="y", alpha=0.3)
    plt.show()

@torch.no_grad()
def visualize_attention(prompt, layer_idx=-1, head_idx=0):
    model.eval()
    ids = tokenize(prompt, add_bos=True, add_eos=False)
```

```
    x = torch.tensor([ids], dtype=torch.long, device=device)
    _ = model(x)

    attn = model.blocks[layer_idx].attn.last_attn[0, head_idx].cpu().numpy()  # (T,
    tokens = [itos[i] for i in x[0].tolist()]

    plt.figure(figsize=(6, 5))
    plt.imshow(attn, aspect="auto", cmap="plasma")
    plt.xticks(range(len(tokens)), tokens, rotation=45)
    plt.yticks(range(len(tokens)), tokens)
    plt.xlabel("Key position")
    plt.ylabel("Query position")
    plt.title(f"Attention map | layer {layer_idx} | head {head_idx}")
    plt.colorbar()
    plt.tight_layout()
    plt.show()

def plot_embedding_pca():
    # simple PCA with numpy
    emb = model.token_emb.weight.detach().cpu().numpy()  # (V, D)
    emb_centered = emb - emb.mean(axis=0, keepdims=True)
    U, S, Vt = np.linalg.svd(emb_centered, full_matrices=False)
    coords = emb_centered @ Vt[:2].T  # (V,2)

    plt.figure(figsize=(7, 6))
    plt.scatter(coords[:, 0], coords[:, 1])
    for i, tok in enumerate(vocab):
        plt.text(coords[i, 0], coords[i, 1], tok, fontsize=9)
    plt.title("Token embedding PCA projection")
    plt.xlabel("PC1")
    plt.ylabel("PC2")
    plt.grid(alpha=0.3)
    plt.show()

@torch.no_grad()
def compare_prompts(prompts, max_new_tokens=10):
    print("\nGenerated samples:")
    for p in prompts:
        out = generate(p, max_new_tokens=max_new_tokens, temperature=0.9, top_k=5)
        print(f"Prompt: {p}")
        print(f"Output: {out}")
        print("-" * 60)

# ----------------------------
# Outputs
# ----------------------------
plot_loss(loss_history)
```



Training shows clear but modest convergence. The loss decreases from about 3.41 at the start to around 2.82 by the end, and the curve drops quickly early on before flattening into a noisy plateau near 2.8–2.85. Overall the performance shows successful learning on the 20k-sequence UniRef50 subset using a lightweight model on CPU but could be optimized further.

(B)

In [4]:
```
# Part (b): 5-sample generation matrix + FASTA output
# Assumes Part (a) cells already ran in this kernel (model, stoi/itos, BOS/EOS, max_

from pathlib import Path

# Reproducible sampling
torch.manual_seed(7)
random.seed(7)
np.random.seed(7)

valid_aas = sorted([tok for tok in stoi.keys() if len(tok) == 1 and tok.isalpha()])
valid_aa_set = set(valid_aas)
print("Valid amino-acid tokens in vocab:", "".join(valid_aas))


def clean_sequence(seq: str) -> str:
    return "".join([ch for ch in seq if ch in valid_aa_set])


@torch.no_grad()
def sample_sequence(prompt: str, max_new_tokens: int, temperature: float = 1.0, top
    model.eval()
```

```
    # Keep only alphabetic chars from prompt, then tokenize via existing pipeline
    prompt = clean_sequence(prompt.upper())
    ids = tokenize(prompt, add_bos=True, add_eos=False)
    x = torch.tensor([ids], dtype=torch.long, device=device)

    # Safety: model positional encoding supports sequence length up to max_len
    max_steps = min(max_new_tokens, (max_len - 1) - x.shape[1])
    max_steps = max(0, max_steps)

    for _ in range(max_steps):
        logits = model(x)
        next_logits = logits[:, -1, :]

        # Avoid sampling special tokens directly
        next_logits[:, PAD] = -1e9
        next_logits[:, BOS] = -1e9

        # Temperature scaling
        next_logits = next_logits / max(temperature, 1e-6)

        # Top-k filtering
        if top_k is not None and top_k > 0:
            k = min(int(top_k), next_logits.shape[-1])
            vals, _ = torch.topk(next_logits, k=k)
            kth = vals[:, -1].unsqueeze(-1)
            next_logits = torch.where(
                next_logits < kth,
                torch.full_like(next_logits, float("-inf")),
                next_logits,
            )

        probs = torch.softmax(next_logits, dim=-1)
        next_token = torch.multinomial(probs, num_samples=1)

        if next_token.item() == EOS:
            break

        x = torch.cat([x, next_token], dim=1)

    seq = detokenize(x[0].tolist(), remove_special=True)
    return clean_sequence(seq)


# Pick 3 natural prefixes (20 aa) from loaded UniRef corpus
natural_prefixes = []
for s in corpus:
    s = clean_sequence(s)
    if len(s) >= 20:
        natural_prefixes.append(s[:20])
    if len(natural_prefixes) == 3:
        break

if len(natural_prefixes) < 3:
    raise RuntimeError("Could not extract three natural 20-aa prefixes from corpus.
```

```
# Agreed 5-sample matrix
experiments = [
    {"id": "S1", "mode": "from_scratch", "prompt": "",                  "temperature
    {"id": "S2", "mode": "from_scratch", "prompt": "",                  "temperature
    {"id": "S3", "mode": "natural_prefix", "prompt": natural_prefixes[0], "temperat
    {"id": "S4", "mode": "natural_prefix", "prompt": natural_prefixes[1], "temperat
    {"id": "S5", "mode": "natural_prefix", "prompt": natural_prefixes[2], "temperat
]


results = []
for cfg in experiments:
    seq = sample_sequence(
        prompt=cfg["prompt"],
        max_new_tokens=cfg["max_new_tokens"],
        temperature=cfg["temperature"],
        top_k=cfg["top_k"],
    )
    results.append({
        **cfg,
        "sequence": seq,
        "length": len(seq),
    })


# Save FASTA
out_dir = Path("outputs") / "problem3"
out_dir.mkdir(parents=True, exist_ok=True)
fasta_path = out_dir / "part_b_5_samples.fasta"

with open(fasta_path, "w", encoding="utf-8") as f:
    for r in results:
        f.write(f">{r['id']}|mode={r['mode']}|temp={r['temperature']}|topk={r['top_
        f.write(r["sequence"] + "\n")


# Print compact summary
print("\nGenerated sequences (Part b):")
for r in results:
    prompt_tag = (r["prompt"][:20] + "...") if len(r["prompt"]) > 20 else r["prompt
    print(
        f"{r['id']}: mode={r['mode']:<14} temp={r['temperature']:<3} "
        f"top_k={r['top_k']:<2} len={r['length']:<3} prompt='{prompt_tag}'"
    )
    print("   ", r["sequence"][:120] + ("..." if len(r["sequence"]) > 120 else ""))

print("\nSaved FASTA:", fasta_path.resolve())
```

```
Valid amino-acid tokens in vocab: ACDEFGHIKLMNPQRSTUVWXY

Generated sequences (Part b):
S1: mode=from_scratch   temp=0.8 top_k=5  len=140 prompt=''
    MSTLSSSRSTSSLSTSSASSATSATATAASTATSTSSSAATSSSTASSSSSASSSSTTASSSPSRTSSRSSASPPTTTSR
SSTRAPSATSSTSAPTPTPSARSSSPTPSSSATSASSSSS...
S2: mode=from_scratch   temp=1.2 top_k=10 len=140 prompt=''
    MRIGSIVAARSPERRTVETSPVTDPPEPPPTPPDEPEPTVADDGVTADRARTSSPATREVVTDDRGAVSTVTVSDGLDVT
RTSAEAVLEDEGVDGRSGRSDEVGERGGAASSGVGGPARA...
S3: mode=natural_prefix temp=0.8 top_k=5  len=140 prompt='MGYIHTALKSAGFHHVIQVD'
    MGYIHTALKSAGFHHVIQVDALRLGAGALLAADGGRVVGAGGAVALAAAAVDVGDGGALGAVAAAGGDADALGVLGVLVA
DGVGDVVGVGAGGVAGVGVVAAAAAAGALDADDVDVAVDV...
S4: mode=natural_prefix temp=1.0 top_k=8  len=140 prompt='MAKSNQKSDSKAFVLALNHE'
    MAKSNQKSDSKAFVLALNHEENENEIIEEDNEEKNNNDNLNSNNSETDTDLENETNSITNITTKKSKTSDSDINKSNSNI
NLNSKENNDNNSINTKSSTDEINENLIDISSKDTTEKSKE...
S5: mode=natural_prefix temp=1.3 top_k=12 len=140 prompt='MRCAKHSLYGFAVPFFSQKG'
    MRCAKHSLYGFAVPFFSQKGPSVTGIIAVGDIRLPLRDPPTPVRRELISREEETRGAGTEVDVSVTAAATAAPARPPAGR
ATASGQARTGTLTQPEGQGRLATLIEQGPSRASVGELRSR...

Saved FASTA: C:\Users\mpave\OneDrive - Massachusetts Institute of Technology\2.174 M
echanics and Materials via Machine Learning\PS3\outputs\problem3\part_b_5_samples.fa
sta
```

I generated five de novo protein candidates using different sampling settings such as from-scratch vs natural-prefix prompting with varied temperature/top-k. The goal was to evaluate how sampling strategy affects sequence diversity and predicted foldability.

Overall the experiments showed trade-off between coherence and creativity. Lower-temperature sampling produced more conservative and repetitive sequences. Higher-temperature sampling increased novelty. Natural-prefix prompting preserved characteristics from the seed sequence while from-scratch generation explored a wider sequence space.

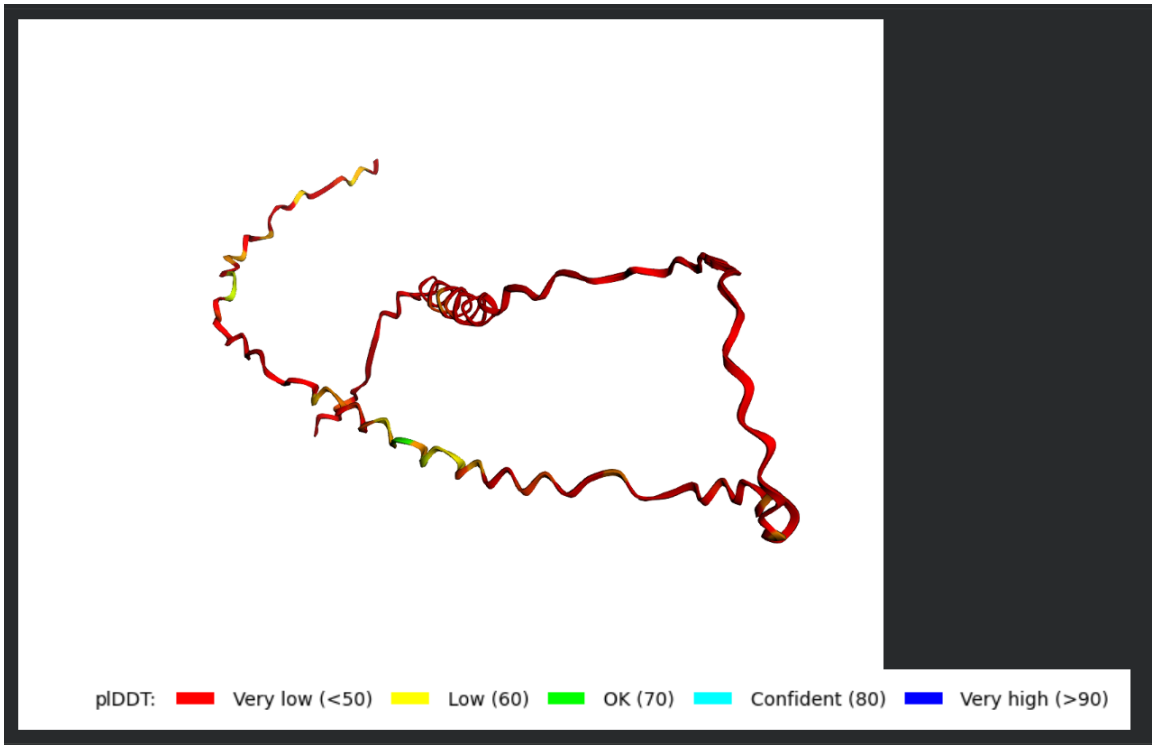
The outputs of my proteins are below:

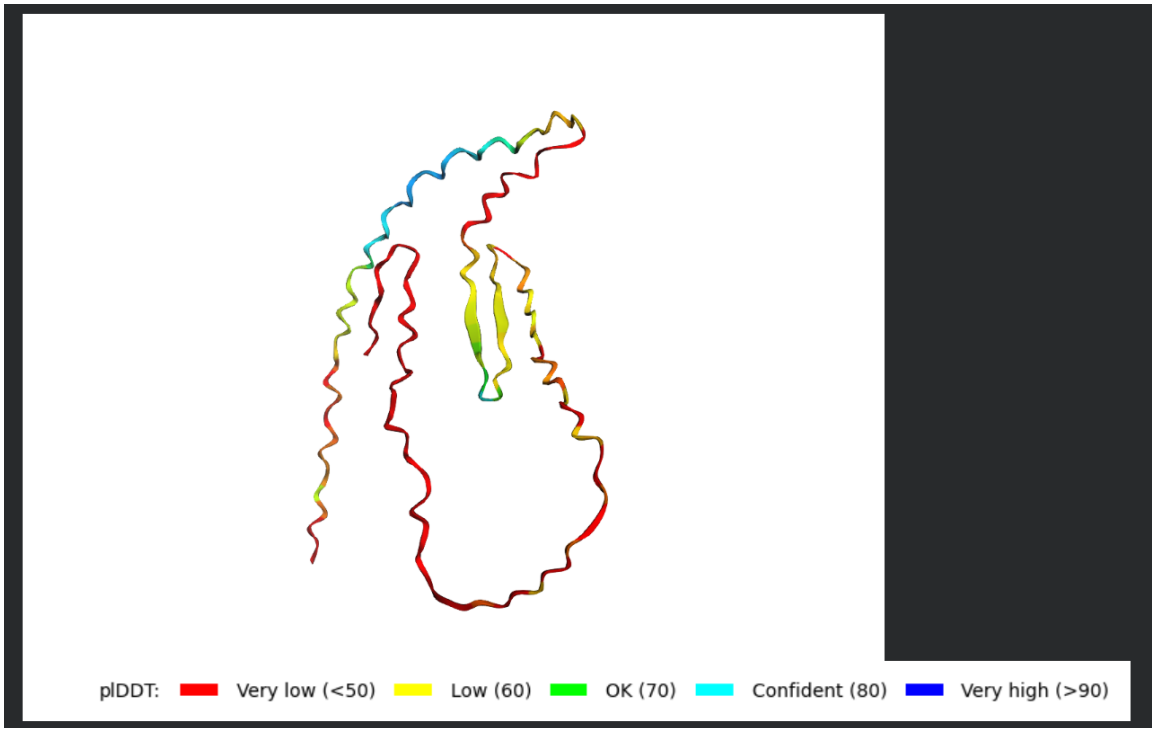
- chain1: mean pLDDT ~ 45.9, pTM ~ 0.17
- chain2: mean pLDDT ~ 54.1, pTM ~ 0.16
- chain3: mean pLDDT ~ 35.0, pTM ~ 0.18
- chain4: mean pLDDT ~ 57.4, pTM ~ 0.13
- chain5: mean pLDDT ~ 42.0, pTM ~ 0.21

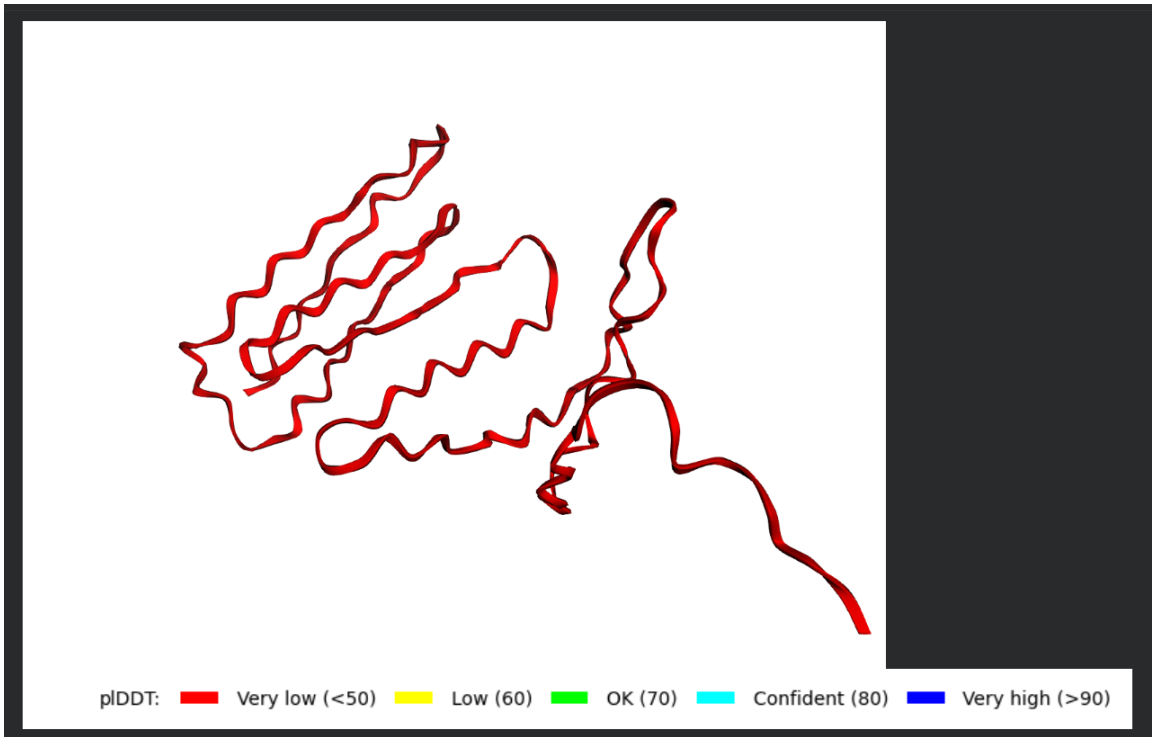
From these values and the structure images below, predicted confidence is generally low-to-moderate which is expected for short training runs on a small GPT trained from scratch on a subset of UniRef50. Among the five samples chain4 and chain2 appear most promising by average pLDDT and chain3 appears least confident.

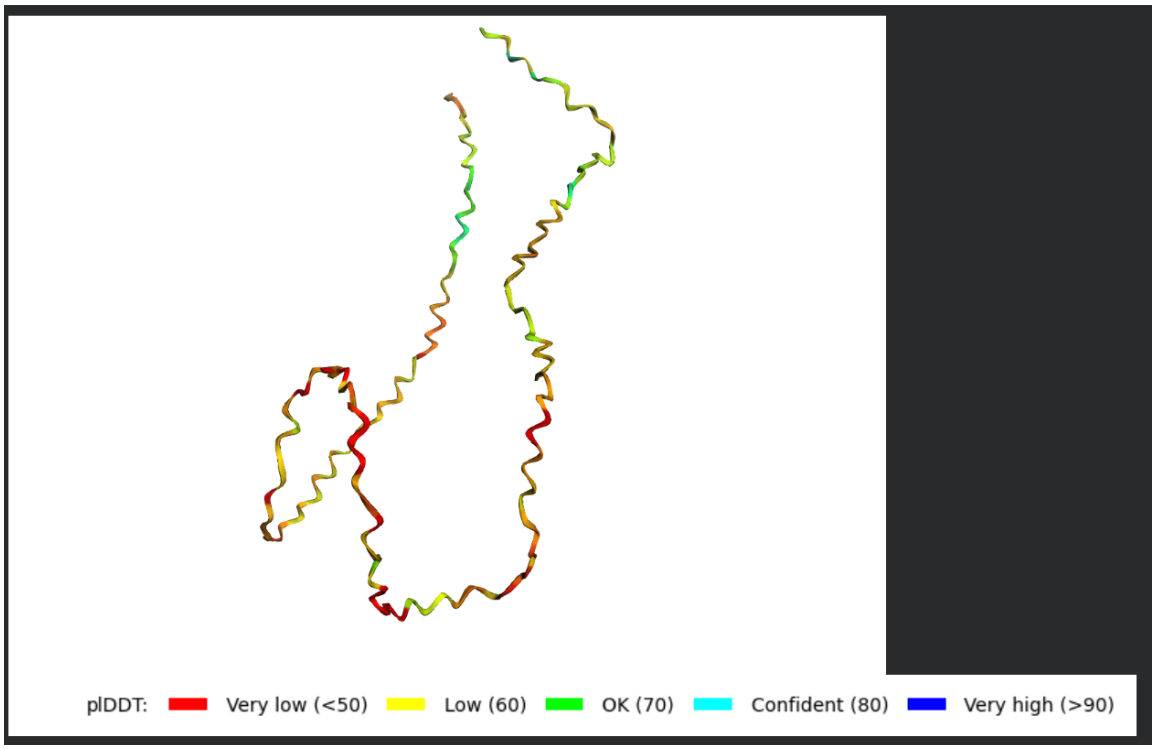
The model can generate diverse protein-like sequences and sampling controls this diversity meaningfully. However, fold confidence remains limited, indicating that stronger models and longer training would likely be needed for robustly designable proteins.

Generated protein structures (chain1-chain4)


plDDT:
Very low (<50)
Low (60)
OK (70)
Confident (80)
Very high (>90)


plDDT:
Very low (<50)
Low (60)
OK (70)
Confident (80)
Very high (>90)





Problem 4

(B)

Case 1

The displacement is fixed to zero at both ends for all time. Physically both endpoints are clamped in place, so the medium cannot move vertically at the boundaries.

Case 2:
The spatial slope is zero at both ends for all time. Physically each endpoint has no tilt.

So, Case 1 enforces fixed displacement at boundaries and Case 2 enforces zero boundary slope.

(A) & (C)

In [16]:
```
# Problem 4 revised (training cell): style aligned with provided PINN notebook
from typing import Callable
from functools import partial

import numpy as np
import matplotlib.pyplot as plt
import torch
from torch import nn
from tqdm.auto import tqdm

# Device
if torch.cuda.is_available():
    device = torch.device("cuda")
elif getattr(torch.backends, "mps", None) and torch.backends.mps.is_available():
    device = torch.device("mps")
else:
    device = torch.device("cpu")
print("Using device:", device)


class PINN(nn.Module):
    """Simple neural network taking (x,t) and predicting u(x,t)."""

    def __init__(self, num_hidden: int, dim_hidden: int, act=nn.Tanh()):
        super().__init__()
        self.layer_in = nn.Linear(2, dim_hidden)
        self.layer_out = nn.Linear(dim_hidden, 1)
        self.middle_layers = nn.ModuleList(
            [nn.Linear(dim_hidden, dim_hidden) for _ in range(num_hidden - 1)]
        )
        self.act = act

    def forward(self, x, t):
        x_stack = torch.cat([x, t], dim=1)
        out = self.act(self.layer_in(x_stack))
        for layer in self.middle_layers:
            out = self.act(layer(out))
        return self.layer_out(out)


def f(nn_approximator: PINN, x: torch.Tensor, t: torch.Tensor) -> torch.Tensor:
    return nn_approximator(x, t)


def df(output: torch.Tensor, input_tensor: torch.Tensor, order: int = 1) -> torch.T
    df_value = output
```

```
    for _ in range(order):
        df_value = torch.autograd.grad(
            df_value,
            input_tensor,
            grad_outputs=torch.ones_like(df_value),
            create_graph=True,
            retain_graph=True,
        )[0]
    return df_value


def dfdt(nn_approximator: PINN, x: torch.Tensor, t: torch.Tensor, order: int = 1):
    return df(f(nn_approximator, x, t), t, order=order)


def dfdx(nn_approximator: PINN, x: torch.Tensor, t: torch.Tensor, order: int = 1):
    return df(f(nn_approximator, x, t), x, order=order)


# Domain and constants
LENGTH = 1.0
TOTAL_TIME = 1.0
C = 1.0

# Loss weights (help enforce BC/IC amplitude and boundary behavior)
W_PDE = 1.0
W_BC = 2.0
W_IC = 5.0

x_domain = [0.0, LENGTH]
t_domain = [0.0, TOTAL_TIME]

n_points_x = 120
n_points_t = 120

x_raw = torch.linspace(x_domain[0], x_domain[1], steps=n_points_x, device=device)
t_raw = torch.linspace(t_domain[0], t_domain[1], steps=n_points_t, device=device)
grids = torch.meshgrid(x_raw, t_raw, indexing="ij")

x_train = grids[0].flatten().reshape(-1, 1).clone().detach().requires_grad_(True)
t_train = grids[1].flatten().reshape(-1, 1).clone().detach().requires_grad_(True)

print("Train grid:", x_train.shape, t_train.shape)


# Case definitions from Problem 4(a)
def initial_condition_case1(x) -> torch.Tensor:
    return torch.sin(4.0 * np.pi * x).reshape(-1, 1)


def initial_velocity_case1(x) -> torch.Tensor:
    return torch.zeros_like(x).reshape(-1, 1)


def initial_condition_case2(x) -> torch.Tensor:
    return torch.cos(4.0 * np.pi * x).reshape(-1, 1)
```

```
def initial_velocity_case2(x) -> torch.Tensor:
    return torch.cos(4.0 * np.pi * x).reshape(-1, 1)


def compute_loss_case1(nn_approximator: PINN, x: torch.Tensor, t: torch.Tensor) ->
    # PDE: u_tt = c^2 u_xx  -> d2u/dx2 - (1/c^2) d2u/dt2 = 0
    interior_loss = dfdx(nn_approximator, x, t, order=2) - (1.0 / C**2) * dfdt(nn_a

    # Dirichlet BCs: u(0,t)=0, u(L,t)=0
    t_vals = torch.unique(t.detach()).reshape(-1, 1).to(device).requires_grad_(True
    boundary_x0 = torch.zeros_like(t_vals, device=device, requires_grad=True)
    boundary_xL = torch.full_like(t_vals, LENGTH, device=device, requires_grad=True
    boundary_loss_x0 = f(nn_approximator, boundary_x0, t_vals)
    boundary_loss_xL = f(nn_approximator, boundary_xL, t_vals)

    # ICs: u(x,0)=sin(4pi x), u_t(x,0)=0
    x_vals = torch.unique(x.detach()).reshape(-1, 1).to(device).requires_grad_(True
    t0 = torch.zeros_like(x_vals, device=device, requires_grad=True)

    f_initial = initial_condition_case1(x_vals)
    ut_initial = initial_velocity_case1(x_vals)

    initial_loss_f = f(nn_approximator, x_vals, t0) - f_initial
    initial_loss_df = dfdt(nn_approximator, x_vals, t0, order=1) - ut_initial

    final_loss = (
        W_PDE * interior_loss.pow(2).mean()
        + W_BC * (boundary_loss_x0.pow(2).mean() + boundary_loss_xL.pow(2).mean())
        + W_IC * (initial_loss_f.pow(2).mean() + initial_loss_df.pow(2).mean())
    )
    return final_loss


def compute_loss_case2(nn_approximator: PINN, x: torch.Tensor, t: torch.Tensor) ->
    # PDE: u_tt = c^2 u_xx
    interior_loss = dfdx(nn_approximator, x, t, order=2) - (1.0 / C**2) * dfdt(nn_a

    # Neumann BCs: u_x(0,t)=0, u_x(L,t)=0
    t_vals = torch.unique(t.detach()).reshape(-1, 1).to(device).requires_grad_(True
    boundary_x0 = torch.zeros_like(t_vals, device=device, requires_grad=True)
    boundary_xL = torch.full_like(t_vals, LENGTH, device=device, requires_grad=True

    boundary_ux0 = dfdx(nn_approximator, boundary_x0, t_vals, order=1)
    boundary_uxL = dfdx(nn_approximator, boundary_xL, t_vals, order=1)

    # ICs: u(x,0)=cos(4pi x), u_t(x,0)=cos(4pi x)
    x_vals = torch.unique(x.detach()).reshape(-1, 1).to(device).requires_grad_(True
    t0 = torch.zeros_like(x_vals, device=device, requires_grad=True)

    f_initial = initial_condition_case2(x_vals)
    ut_initial = initial_velocity_case2(x_vals)

    initial_loss_f = f(nn_approximator, x_vals, t0) - f_initial
    initial_loss_df = dfdt(nn_approximator, x_vals, t0, order=1) - ut_initial
```

```
    final_loss = (
        W_PDE * interior_loss.pow(2).mean()
        + W_BC * (boundary_ux0.pow(2).mean() + boundary_uxL.pow(2).mean())
        + W_IC * (initial_loss_f.pow(2).mean() + initial_loss_df.pow(2).mean())
    )
    return final_loss


def train_model(
    nn_approximator: PINN,
    loss_fn: Callable,
    learning_rate: float = 2e-4,
    max_epochs: int = 4000,
    iprint: int = 100,
    use_lbfgs: bool = True,
    lbfgs_steps: int = 200,
):
    losslist = []
    optimizer = torch.optim.Adam(nn_approximator.parameters(), lr=learning_rate)

    for epoch in tqdm(range(max_epochs)):
        try:
            loss = loss_fn(nn_approximator)
            optimizer.zero_grad()
            loss.backward()
            optimizer.step()

            losslist.append(loss.item())
            if epoch % iprint == 0 or epoch == max_epochs - 1:
                print(f"Epoch: {epoch:4d} - Loss: {float(loss):.6f}")
        except KeyboardInterrupt:
            break

    # PINN refinement stage (often improves fit significantly)
    if use_lbfgs:
        print(f"Starting LBFGS refinement for {lbfgs_steps} steps...")
        lbfgs = torch.optim.LBFGS(
            nn_approximator.parameters(),
            lr=1.0,
            max_iter=lbfgs_steps,
            history_size=50,
            line_search_fn="strong_wolfe",
        )

        def closure():
            lbfgs.zero_grad()
            l = loss_fn(nn_approximator)
            l.backward()
            return l

        lbfgs.step(closure)
        loss_after = float(loss_fn(nn_approximator).item())
        losslist.append(loss_after)
        print(f"LBFGS final loss: {loss_after:.6f}")
```

```
    return nn_approximator, losslist


# Train both cases
loss_fn_case1 = partial(compute_loss_case1, x=x_train, t=t_train)
loss_fn_case2 = partial(compute_loss_case2, x=x_train, t=t_train)

model_case1_rev = PINN(num_hidden=5, dim_hidden=96).to(device)
model_case2_rev = PINN(num_hidden=5, dim_hidden=96).to(device)

print("\nTraining revised Case 1 model...")
model_case1_rev, loss_case1_rev = train_model(
    model_case1_rev,
    loss_fn_case1,
    learning_rate=2e-4,
    max_epochs=4000,
    iprint=200,
    use_lbfgs=True,
    lbfgs_steps=200,
)

print("\nTraining revised Case 2 model...")
model_case2_rev, loss_case2_rev = train_model(
    model_case2_rev,
    loss_fn_case2,
    learning_rate=2e-4,
    max_epochs=4000,
    iprint=200,
    use_lbfgs=True,
    lbfgs_steps=200,
)

print("\nRevised training complete. Run next cell for plots + movies.")
```

```
Using device: cpu
Train grid: torch.Size([14400, 1]) torch.Size([14400, 1])

Training revised Case 1 model...
  0%|          | 0/4000 [00:00<?, ?it/s]
  0%|          | 1/4000 [00:00<28:34,  2.33it/s]
Epoch:    0 - Loss: 2.509147
  5%|▌         | 201/4000 [00:51<17:38,  3.59it/s]
Epoch:  200 - Loss: 2.299661
 10%|▉         | 401/4000 [01:43<15:18,  3.92it/s]
Epoch:  400 - Loss: 2.293288
 15%|▉         | 601/4000 [02:35<12:01,  4.71it/s]
Epoch:  600 - Loss: 2.283805
 20%|▉         | 801/4000 [03:18<11:21,  4.69it/s]
Epoch:  800 - Loss: 2.262624
 25%|▉         | 1001/4000 [04:02<10:44,  4.65it/s]
Epoch: 1000 - Loss: 2.255944
 30%|▉         | 1201/4000 [04:47<10:07,  4.60it/s]
Epoch: 1200 - Loss: 2.258029
 35%|▉         | 1401/4000 [05:32<09:27,  4.58it/s]
Epoch: 1400 - Loss: 2.250914
 40%|▉         | 1601/4000 [06:17<09:38,  4.15it/s]
```

```
Epoch: 1600 - Loss: 2.247282
 45%|▌▌▌▌      | 1801/4000 [07:04<09:12,  3.98it/s]
Epoch: 1800 - Loss: 2.239137
 50%|▌▌▌▌▌     | 2001/4000 [07:51<07:18,  4.55it/s]
Epoch: 2000 - Loss: 2.211200
 55%|▌▌▌▌▌▌    | 2201/4000 [08:36<06:31,  4.59it/s]
Epoch: 2200 - Loss: 2.112028
 60%|▌▌▌▌▌▌    | 2401/4000 [09:21<05:45,  4.62it/s]
Epoch: 2400 - Loss: 1.923009
 65%|▌▌▌▌▌▌▌   | 2601/4000 [10:09<06:00,  3.89it/s]
Epoch: 2600 - Loss: 1.757014
 70%|▌▌▌▌▌▌▌▌  | 2801/4000 [11:01<05:04,  3.93it/s]
Epoch: 2800 - Loss: 1.624460
 75%|▌▌▌▌▌▌▌▌  | 3001/4000 [11:52<04:11,  3.97it/s]
Epoch: 3000 - Loss: 1.473717
 80%|▌▌▌▌▌▌▌▌▌ | 3201/4000 [12:44<03:32,  3.75it/s]
Epoch: 3200 - Loss: 1.355304
 85%|▌▌▌▌▌▌▌▌▌ | 3401/4000 [13:35<02:28,  4.03it/s]
Epoch: 3400 - Loss: 1.255632
 90%|▌▌▌▌▌▌▌▌▌▌| 3601/4000 [14:27<01:43,  3.86it/s]
Epoch: 3600 - Loss: 1.179016
 95%|▌▌▌▌▌▌▌▌▌▌| 3801/4000 [15:20<00:55,  3.59it/s]
Epoch: 3800 - Loss: 1.113740
100%|▌▌▌▌▌▌▌▌▌▌| 4000/4000 [16:12<00:00,  4.11it/s]
Epoch: 3999 - Loss: 1.053695
Starting LBFGS refinement for 200 steps...
LBFGS final loss: 0.666193

Training revised Case 2 model...
  0%|          | 1/4000 [00:00<18:18,  3.64it/s]
Epoch:    0 - Loss: 5.049925
  5%|▌         | 201/4000 [00:53<16:32,  3.83it/s]
Epoch:  200 - Loss: 4.988974
 10%|▌         | 401/4000 [01:46<15:38,  3.84it/s]
Epoch:  400 - Loss: 4.892314
 15%|▌         | 601/4000 [02:39<15:02,  3.77it/s]
Epoch:  600 - Loss: 3.778712
 20%|▌         | 801/4000 [03:32<14:44,  3.62it/s]
Epoch:  800 - Loss: 3.530515
 25%|▌▌        | 1001/4000 [04:26<13:06,  3.81it/s]
Epoch: 1000 - Loss: 3.442465
 30%|▌▌        | 1201/4000 [05:20<12:23,  3.76it/s]
Epoch: 1200 - Loss: 3.374214
 35%|▌▌▌       | 1401/4000 [06:10<10:04,  4.30it/s]
Epoch: 1400 - Loss: 3.329572
 40%|▌▌▌       | 1601/4000 [07:03<09:54,  4.03it/s]
Epoch: 1600 - Loss: 3.289018
 45%|▌▌▌▌      | 1801/4000 [08:05<11:02,  3.32it/s]
Epoch: 1800 - Loss: 3.252002
 50%|▌▌▌▌▌     | 2001/4000 [09:05<07:59,  4.17it/s]
Epoch: 2000 - Loss: 3.221528
 55%|▌▌▌▌▌▌    | 2201/4000 [10:02<07:52,  3.81it/s]
Epoch: 2200 - Loss: 3.197811
 60%|▌▌▌▌▌▌    | 2401/4000 [11:00<08:23,  3.18it/s]
Epoch: 2400 - Loss: 3.266760
 65%|▌▌▌▌▌▌▌   | 2601/4000 [12:01<06:58,  3.35it/s]
```

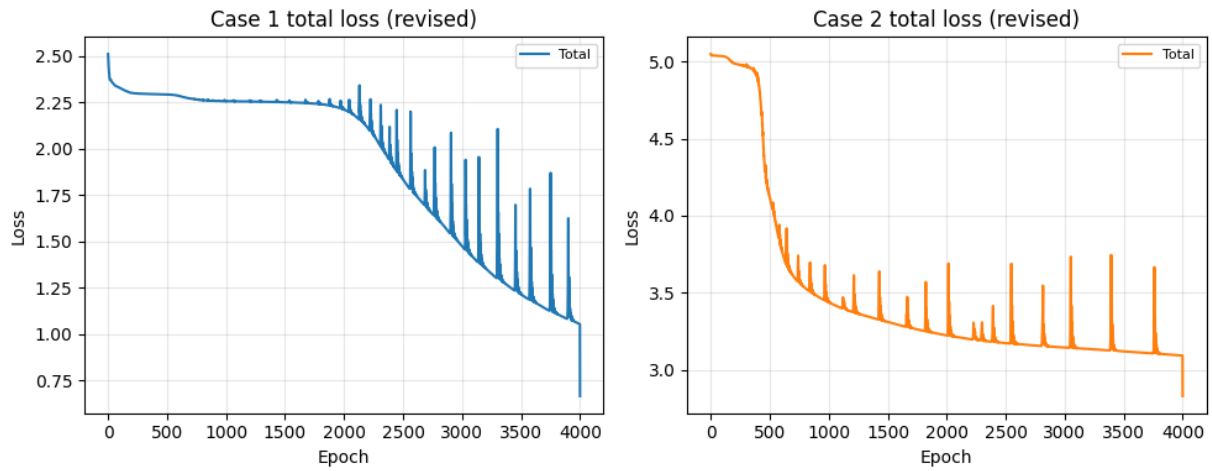
```
Epoch: 2600 - Loss: 3.167696
 70%|▊▊▊▊▊▊▊   | 2801/4000 [13:01<06:37,  3.01it/s]
Epoch: 2800 - Loss: 3.153302
 75%|▊▊▊▊▊▊▊▊  | 3001/4000 [14:04<05:16,  3.16it/s]
Epoch: 3000 - Loss: 3.143102
 80%|▊▊▊▊▊▊▊▊▊ | 3201/4000 [15:08<04:30,  2.96it/s]
Epoch: 3200 - Loss: 3.133863
 85%|▊▊▊▊▊▊▊▊▊▊| 3401/4000 [16:11<03:06,  3.21it/s]
Epoch: 3400 - Loss: 3.223529
 90%|▊▊▊▊▊▊▊▊▊▊| 3601/4000 [17:14<02:08,  3.10it/s]
Epoch: 3600 - Loss: 3.113263
 95%|▊▊▊▊▊▊▊▊▊▊| 3801/4000 [18:18<01:03,  3.11it/s]
Epoch: 3800 - Loss: 3.103895
100%|▊▊▊▊▊▊▊▊▊▊| 4000/4000 [19:27<00:00,  3.43it/s]
Epoch: 3999 - Loss: 3.090951
Starting LBFGS refinement for 200 steps...
LBFGS final loss: 2.828388

Revised training complete. Run next cell for plots + movies.
```

In [17]:

```python
# Problem 4 revised (plotting cell): total loss + snapshots + movies
from pathlib import Path
from matplotlib.animation import FuncAnimation, PillowWriter

out_dir = Path("charts") / "problem4" / "revised_like_given"
out_dir.mkdir(parents=True, exist_ok=True)

# Total loss only (as requested)
plt.figure(figsize=(10, 4))
plt.subplot(1, 2, 1)
plt.plot(loss_case1_rev, color="C0", label="Total")
plt.title("Case 1 total loss (revised)")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.grid(alpha=0.3)
plt.legend(fontsize=8)

plt.subplot(1, 2, 2)
plt.plot(loss_case2_rev, color="C1", label="Total")
plt.title("Case 2 total loss (revised)")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.grid(alpha=0.3)
plt.legend(fontsize=8)

plt.tight_layout()
loss_path = out_dir / "part4_revised_total_loss.png"
plt.savefig(loss_path, dpi=150)
plt.show()
print("Saved:", loss_path.resolve())


@torch.no_grad()
def predict_profile(nn_trained: PINN, x_values: np.ndarray, t_value: float):
    x_t = torch.tensor(x_values.reshape(-1, 1), dtype=torch.float32, device=device)
    t_t = torch.full_like(x_t, float(t_value), device=device)
```

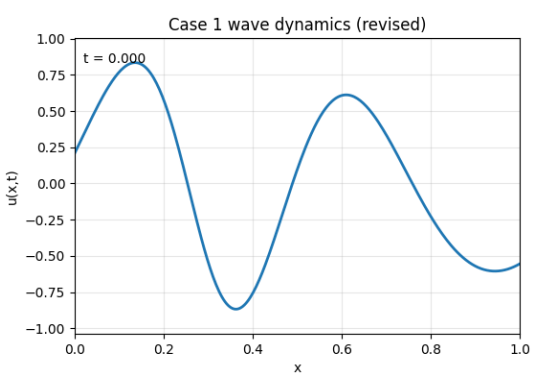
```
    y = f(nn_trained, x_t, t_t)
    return y.detach().cpu().numpy().reshape(-1)


# Use a shared x-grid for GIFs
x_plot = np.linspace(0.0, LENGTH, 300)


def save_wave_gif(nn_trained: PINN, title: str, out_path: Path, n_frames: int = 90)
    fig, ax = plt.subplots(figsize=(6, 4))
    line, = ax.plot([], [], lw=2)
    txt = ax.text(0.02, 0.95, "", transform=ax.transAxes, va="top")

    t_vals = np.linspace(0.0, TOTAL_TIME, n_frames)
    y_all = [predict_profile(nn_trained, x_plot, tt) for tt in t_vals]
    y_min = min(np.min(y) for y in y_all)
    y_max = max(np.max(y) for y in y_all)
    pad = 0.1 * (y_max - y_min + 1e-6)

    ax.set_xlim(0.0, LENGTH)
    ax.set_ylim(y_min - pad, y_max + pad)
    ax.set_xlabel("x")
    ax.set_ylabel("u(x,t)")
    ax.set_title(title)
    ax.grid(alpha=0.3)

    def init():
        line.set_data([], [])
        txt.set_text("")
        return line, txt

    def animate(i):
        line.set_data(x_plot, y_all[i])
        txt.set_text(f"t = {t_vals[i]:.3f}")
        return line, txt

    ani = FuncAnimation(fig, animate, frames=n_frames, init_func=init, interval=80,
    ani.save(out_path, writer=PillowWriter(fps=15))
    plt.close(fig)
    print("Saved:", out_path.resolve())


case1_gif = out_dir / "case1_wave_revised.gif"
case2_gif = out_dir / "case2_wave_revised.gif"

save_wave_gif(model_case1_rev, "Case 1 wave dynamics (revised)", case1_gif)
save_wave_gif(model_case2_rev, "Case 2 wave dynamics (revised)", case2_gif)

from IPython.display import HTML, display

display(HTML(f"""
<div style='display:flex; gap:16px; align-items:flex-start;'>
  <div>
    <div><b>Case 1 wave GIF</b></div>
    <img src='{case1_gif.as_posix()}' width='420'>
  </div>
```

```
  <div>
    <div><b>Case 2 wave GIF</b></div>
    <img src='{case2_gif.as_posix()}' width='420'>
  </div>
</div>
"""))

print("\nRevised plotting complete: total-loss plot and GIFs saved/displayed.")
```

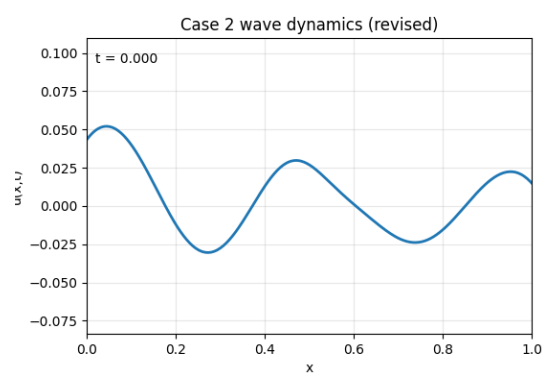



```
Saved: C:\Users\mpave\OneDrive - Massachusetts Institute of Technology\2.174 Mechani
cs and Materials via Machine Learning\PS3\charts\problem4\revised_like_given\part4_r
evised_total_loss.png
Saved: C:\Users\mpave\OneDrive - Massachusetts Institute of Technology\2.174 Mechani
cs and Materials via Machine Learning\PS3\charts\problem4\revised_like_given\case1_w
ave_revised.gif
Saved: C:\Users\mpave\OneDrive - Massachusetts Institute of Technology\2.174 Mechani
cs and Materials via Machine Learning\PS3\charts\problem4\revised_like_given\case2_w
ave_revised.gif
```

Case 1 wave GIF



Case 2 wave GIF



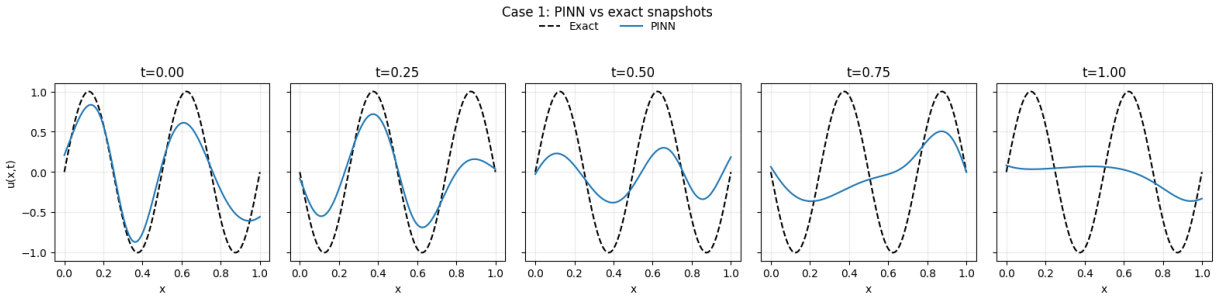

```
Revised plotting complete: total-loss plot and GIFs saved/displayed.
```

(D)

In [18]:
```
# Problem 4(d): quantitative comparison to exact Case-1 solution
import numpy as np
import matplotlib.pyplot as plt
import torch
from pathlib import Path
```

```
# Requires model_case1_rev from the revised Part (c) training cell.
if "model_case1_rev" not in globals():
    raise RuntimeError("Run the revised Part (c) training cell first (model_case1_r


@torch.no_grad()
def predict_case1_pinn(x_values: np.ndarray, t_value: float):
    x_t = torch.tensor(x_values.reshape(-1, 1), dtype=torch.float32, device=device)
    t_t = torch.full_like(x_t, float(t_value))
    y = f(model_case1_rev, x_t, t_t)
    return y.detach().cpu().numpy().reshape(-1)


def exact_case1(x_values: np.ndarray, t_value: float, c_val: float = 1.0):
    return np.sin(4.0 * np.pi * x_values) * np.cos(4.0 * np.pi * c_val * t_value)


out_dir = Path("charts") / "problem4" / "revised_like_given"
out_dir.mkdir(parents=True, exist_ok=True)

# --- Snapshot overlay plot ---
x_plot = np.linspace(0.0, LENGTH, 300)
t_snap = [0.0, 0.25 * TOTAL_TIME, 0.5 * TOTAL_TIME, 0.75 * TOTAL_TIME, 1.0 * TOTAL_

fig, axes = plt.subplots(1, len(t_snap), figsize=(16, 3.8), sharey=True)
for ax, tt in zip(axes, t_snap):
    u_pred = predict_case1_pinn(x_plot, tt)
    u_ex = exact_case1(x_plot, tt, c_val=C)
    ax.plot(x_plot, u_ex, "k--", lw=1.5, label="Exact")
    ax.plot(x_plot, u_pred, "C0", lw=1.5, label="PINN")
    ax.set_title(f"t={tt:.2f}")
    ax.set_xlabel("x")
    ax.grid(alpha=0.25)
axes[0].set_ylabel("u(x,t)")
handles, labels = axes[0].get_legend_handles_labels()
fig.suptitle("Case 1: PINN vs exact snapshots", y=1.02)
fig.legend(handles, labels, loc="upper center", bbox_to_anchor=(0.5, 1.00), ncol=2,
fig.tight_layout(rect=[0, 0, 1, 0.94])
compare_path = out_dir / "part4_case1_exact_vs_pinn_snapshots.png"
fig.savefig(compare_path, dpi=150, bbox_inches="tight")
plt.show()
print("Saved:", compare_path.resolve())

# --- Space-time error metrics ---
x_eval = np.linspace(0.0, LENGTH, 200)
t_eval = np.linspace(0.0, TOTAL_TIME, 120)

U_pred = np.zeros((len(t_eval), len(x_eval)))
U_exact = np.zeros_like(U_pred)
for i, tt in enumerate(t_eval):
    U_pred[i, :] = predict_case1_pinn(x_eval, tt)
    U_exact[i, :] = exact_case1(x_eval, tt, c_val=C)

E = U_pred - U_exact
mse = float(np.mean(E**2))
rmse = float(np.sqrt(mse))
```

```
mae = float(np.mean(np.abs(E)))
max_abs = float(np.max(np.abs(E)))

print(f"Case 1 error metrics: MSE={mse:.6e}, RMSE={rmse:.6e}, MAE={mae:.6e}, Max|er
```




```
Saved: C:\Users\mpave\OneDrive - Massachusetts Institute of Technology\2.174 Mechani
cs and Materials via Machine Learning\PS3\charts\problem4\revised_like_given\part4_c
ase1_exact_vs_pinn_snapshots.png
Case 1 error metrics: MSE=1.639169e-01, RMSE=4.048665e-01, MAE=3.229502e-01, Max|er
r|=1.231487e+00
```

For Case 1, the exact solution is $u(x,t) = \sin(4\pi x)\cos(4\pi t)$

The PINN prediction agrees with the theoretical Case 1 behavior in the main trend, especially at earlier times, although there is still some mismatch in amplitude and phase later in time. This means the wave shape stays in fixed positions and only oscillates in time. So this is standing-wave behavior, not a traveling wave that propagates left or right across the domain.